

CO769 Optimal Transport

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1 Preliminaries

1.1 Convex Sets

See my notes on Pmath753 Functional Analysis for more on convex sets.

Definition 1.1. Let X be a vector space, a set $K \subseteq X$ is **convex** if

$$\forall x, y \in K, t \in (0, 1), (1 - t)x + ty \in K.$$

Definition 1.2. Let X be a vector space, and $A \subseteq X$. The **convex hull** of A is

$$\text{conv}(A) := \left\{ x = \sum_{i=1}^n \lambda_i x_i \mid n \in \mathbb{N}, x_i \in A, \lambda_i \geq 0, \sum_{i=1}^n \lambda_i = 1 \right\}.$$

The elements in $\text{conv}(A)$ are called **convex combinations** of A .

Proposition 1.1. *The convex hull is the smallest convex set containing A .*

Definition 1.3. Let X be a vector space, and $A \subseteq X$. The **affine hull** of A is

$$\text{aff}(A) := \left\{ x = \sum_{i=1}^n \lambda_i x_i \mid n \in \mathbb{N}, x_i \in A, \lambda_i \in \mathbb{R}, \sum_{i=1}^n \lambda_i = 1 \right\}.$$

The elements in $\text{aff}(A)$ are called **affine combinations** of A .

Definition 1.4. Let X be a Vector Space, and let $\emptyset \neq A \subseteq X$ be convex. A **face** of A is some convex $\emptyset \neq F \subseteq A$, such that for all $t \in (0, 1), x, y \in A$, if $(1 - t)x + ty \in F$, then $x, y \in F$.

If a face $F = \{z\}$, we call z an **extreme point** of A . Namely, $\nexists x, y \in A \setminus \{z\}, \lambda \in (0, 1)$, such that $\lambda x + (1 - \lambda)y = z$. i.e., it does not live on any line segment between two other elements.

$\text{Ext}(A)$ is the set of extreme points of A .

Proposition 1.2. *Let X be a Vector Space, and let $\emptyset \neq A \subseteq X$ be convex. Suppose F is a face for A , and F' is a face for F , then F' is also a face for A .*

Proof. Consider any $x, y \in A, t \in (0, 1)$, with $(1 - t)x + ty \in F' \subseteq F$. Since F is a face of A , $x, y \in F$. Since F' is a face of F , $x, y \in F'$. Thus, F' is a face of A . $\square$

Theorem 1.3 (Krein-Milman). *Let $\emptyset \neq K \subseteq \mathbb{R}^d$ be convex and compact, then*

$$K = \text{conv}(\text{Ext}(K)).$$

Theorem 1.4 (Caratheodory). *If $u \in \text{conv} \{x_1, \dots, x_l\} \subseteq \mathbb{R}^d$, with $\dim(\text{aff} \{x_1, \dots, x_l\}) \leq n$, we must have $i_1, \dots, i_{n+1} \in [l]$, such that $u \in \text{conv} \{x_{i_1}, \dots, x_{i_{n+1}}\}$.*

Definition 1.5. Let X be a Vector Space, and let $A \subseteq X$. The **relative interior** of A is

$$\text{relint}(A) := \{x \in A : \exists r > 0, \text{ such that } \bar{B}(x, r) \cap \text{aff}(A) \subseteq A\}.$$

Proposition 1.5. *Let X be a Vector Space, and let $A \subseteq X$ be convex. We have*

$$\begin{aligned} \text{relint}(A) &= \{x \in A : \forall y \in A, \exists \lambda > 0, \text{ such that } \lambda x + (1 - \lambda)y \in A\} \\ &= \{x \in A : \forall y \in A, \exists \lambda \in (0, 1), z \in A, \text{ such that } x = \lambda y + (1 - \lambda)z\}. \end{aligned}$$

1.2 Polyhedrons

Definition 1.6. A (closed) **halfspace** $H \subseteq \mathbb{R}^d$ is $H := \{x \in \mathbb{R}^d : c^T x \leq d\}$ for some $c \in \mathbb{R}^d \setminus \{0\}, d \in \mathbb{R}$.

Definition 1.7. A **hyperplane** $H \subseteq \mathbb{R}^d$ is $H := \{x \in \mathbb{R}^d : c^T x = d\}$ for some $c \in \mathbb{R}^d \setminus \{0\}, d \in \mathbb{R}$.

Definition 1.8. A **polyhedron** $P \subseteq \mathbb{R}^d$ is the intersection of finitely many halfplanes.

Definition 1.9. A **polytope** is a bounded polyhedron.

Definition 1.10. Let P be a polyhedron. A face F of P is **exposed** if $\exists c \in \mathbb{R}^d, d \in \mathbb{R}$, such that

$$\begin{aligned} \forall x \in P \setminus F, \quad c^T \hat{x} &= d \\ \forall x \in F, \quad c^T x &\geq d. \end{aligned}$$

Namely, F is the intersection of P with a hyperplane $H := \{x \in \mathbb{R}^d : c^T x = d\}$ such that $P \subseteq \{x \in \mathbb{R}^d : c^T x \geq d\}$.

Also, we allow $H := \mathbb{R}^d$ so that P is an exposed face of itself.

Definition 1.11. Let P be a polyhedron. $\hat{x} \in P$ is **exposed** if $\exists c \in \mathbb{R}^d$, such that $\forall x \in P \setminus \{\hat{x}\}, c^T \hat{x} < c^T x$. Namely, there is the hyperplane that intersects P only at $\hat{x}$.

Definition 1.12. Let P be a polyhedron defined by the constraints

$$\begin{aligned} c_1^T x &\leq d_1 \\ &\vdots \\ c_n^T x &\leq d_n, \end{aligned}$$

and $\hat{x} \in P$. The **active constraint vectors** of P at $\hat{x}$ are the ones such that equality holds at $\hat{x}$. Namely, $\{c_i : c_i^T \hat{x} = d_i\}$.

Theorem 1.6. Given a polyhedron P and a point $\hat{x}$, the following are equal:

1. $\hat{x}$ is an extreme point;
2. $\hat{x}$ is exposed;
3. The subset of active constraint vectors at $\hat{x}$ spans $\mathbb{R}^d$. Namely, $\dim(\text{Span}(\{c_i : c_i^T \hat{x} = d_i\})) = d$.

In this case, we call $\hat{x}$ an **vertex** of P .

Remark. In the setting of general convex sets, any exposed point is an extreme point, but the converse may not be true.

Theorem 1.7 (Dubbin). Suppose $K \subseteq \mathbb{R}^d$ is a polytope, and $H_1, \dots, H_n$ are hyperplanes. Let $L := K \cap H_1 \cap \dots \cap H_n$, which is also a polytope, then any extreme point of L is a convex combination of at most $n+1$ extreme points of K .

Proof. Pick any extreme point $p \in L$. Let F be the minimal exposed face of K containing p , with $d' := \dim(F)$, then p is in the relative interior of F , otherwise F is not minimal. Thus, there is $r > 0$, such that $\bar{B}(p, r) \cap \text{aff}(F) \subseteq F$.

Let $B^* := \bar{B}(p, r) \cap \text{aff}(F) \cap H_1 \cap \dots \cap H_n \subseteq L$. Since intersecting a hyperplane can reduce the dimension by at most 1, $\dim(B^*) \geq \dim(\bar{B}(p, r) \cap \text{aff}(F)) - n \geq d' - n$.

Since p is an extreme point of L , L cannot contain a ball (intersected by hyperplanes) centred at p of positive dimension. Thus, $d' - n \leq 0$, so $\dim(F) = d' \leq n$. By Caratheodory Theorem, p lies in convex hull of at most $n+1$ extreme points of F , thus of K . $\square$

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1.3 Linear Programming

Definition 1.13. Linear Programming is minimizing or maximizing a linear **objective** function over a polyhedron, the **feasible region**.

Theorem 1.8. *For any liner programming problem, exactly one of the following is true:*

1. *It is infeasible;*
2. *It is unbounded;*
3. *It has a minimizer.*

Theorem 1.9. *Suppose a linear programming problem is feasible and bounded, and the feasible region contains no line, then it has an optimizer at a vertex.*

1.4 Continuous Functions

See my notes on Pmath753 Functional Analysis for more on continuous functions.

Definition 1.14. Let $C(X)$ be the collection of functions $f : X \rightarrow \mathbb{R}$ that are continuous.

Definition 1.15. For $f \in C(X)$, the **support** of it is

$$\text{Supp}(f) := \overline{f^{-1}(\mathbb{R} \setminus \{0\})} \subseteq X.$$

Definition 1.16. The set of **compactly supported functions** are

$$C_c(X) := \{f \in C(X) : \text{Supp}(f) \text{ is compact}\}.$$

Definition 1.17. The set of **bounded functions** are

$$C_b(X) := \left\{f \in C(X) : \|f\|_{C(X)} < \infty\right\},$$

where

$$\|f\|_{C(X)} := \sup_{x \in X} |f(x)|$$

is the **supremum norm**.

Definition 1.18.

$$C_0(X) := \overline{C_c(X)}^{\|\cdot\|_{C(X)}}.$$

Proposition 1.10. *Let $(X, \|\cdot\|)$ be a normed space.*

$$C_0(X) = \{f \in C(X) : \forall \epsilon > 0, \exists R > 0, \text{ such that } \forall \|x\| \geq R, |f(x)| < \epsilon\}$$

is the set of functions that vanishes at infinity.

1.5 Dual Spaces

See my notes on Pmath753 Functional Analysis for more on dual spaces and weak topologies.

Definition 1.19. Let $(X, \|\cdot\|)$ be a normed vector space over $\mathbb{F}$. A **functional** is an operator that maps into $\mathbb{F}$.

Definition 1.20. Let $(X, \|\cdot\|)$ be a normed vector space. The **dual norm** of a linear functional ϕ is defined to be

$$\|\phi\|_{X^*} := \sup_{\|x\| \leq 1} |\phi(x)| = \sup_{x \neq 0 \in X} \frac{|\phi(x)|}{\|x\|}.$$

ϕ is **bounded** if $\|\phi\|_{X^*} < \infty$.

Definition 1.21. Let $(X, \|\cdot\|)$ be a normed vector space over $\mathbb{F}$. The **dual space** of X is the collection of bounded linear functionals on X , denoted

$$X^* := B(X, \mathbb{F}) = \{\phi : X \rightarrow \mathbb{F} : \phi \text{ is linear and bounded}\}.$$

Definition 1.22. Let $(X, \|\cdot\|)$ be a normed vector space over $\mathbb{F}$. For a subspace $Y \subseteq X^*$, we define the **duality pairing** to be $\langle \cdot, \cdot \rangle_{Y, X} : Y \times X \rightarrow \mathbb{F}$ by $\langle \phi, x \rangle_{Y, X} := \phi(x)$, the **action** of ϕ on x .

Definition 1.23. Let $(X, \|\cdot\|)$ be a normed vector space. Let $(x_n)_{n=1}^\infty$ be a sequence in X , and $x \in X$. We say $(x_n)_{n=1}^\infty$ **(strongly) converges** to x , denoted $x_n \rightarrow x$, if $\lim_{n \rightarrow \infty} \|x - x_n\| = 0$.

We say $(x_n)_{n=1}^\infty$ **weakly converges** to x , denoted $x_n \rightharpoonup x$, if $\forall \phi \in X^*$, $\phi(x_n) \rightarrow \phi(x)$.

Definition 1.24. Let $(X, \|\cdot\|)$ be a normed vector space. Let $(\phi_n)_{n=1}^\infty$ be a sequence in X^* , and $\phi \in X^*$. We say $(\phi_n)_{n=1}^\infty$ **(strongly) converges** to ϕ , denoted $\phi_n \rightarrow \phi$, if $\lim_{n \rightarrow \infty} \|\phi - \phi_n\|_{X^*} = 0$.

We say $(\phi_n)_{n=1}^\infty$ **weak-* converges** to ϕ , denoted $\phi_n \rightharpoonup^* \phi$, if $\forall x \in X$, $\phi_n(x) \rightarrow \phi(x)$.

Definition 1.25. We say a subset $S \subseteq X$ is dense if $\bar{S} = X$.

Definition 1.26. X is **separable** if there is a countable dense subset.

Example 1.5.1. $C_c(\mathbb{R}^d)$ is separable, since the set of all finite piecewise linear continuous functions with rational coordinates that are zero on both ends is countable and dense in $C_c(\mathbb{R}^d)$. Thus $C_0(\mathbb{R}^d)$ is also separable. On the other hand, $C_b(\mathbb{R}^d)$ is not separable.

Theorem 1.11 (Banach-Alaoglu). *Let $(X, \|\cdot\|)$ be a normed vector space, then the unit ball $\bar{B}^{(X^*, \|\cdot\|)}(0, 1)$ is compact in the weak-* topology.*

Corollary 1.12. *Let $(X, \|\cdot\|)$ be a separable normed vector space, and $(u_k)_{k=1}^\infty \subset X$ be a bounded sequence, then $\exists (u_{k_j})_{j=1}^\infty$ a subsequence, and $u \in X$, such that $u_{k_j} \rightharpoonup u$.*

Theorem 1.13 (Geometric Hahn–Banach Theorem - First Separation). *Let $(X, \|\cdot\|)$ be a normed vector space, $A, B \subseteq X$ be disjoint convex sets. Suppose A is open, then $\exists t \in \mathbb{R}$, $\phi \in X^*$, such that*

$$\forall x \in A, y \in B, \phi(x) < t \leq \phi(y).$$

Namely, $\phi(A), \phi(B)$ can be separated.

1.6 Measure Theory

See my notes on Pmath651 Measure Theory for more on measures.

Definition 1.27. Let X be a set and $\mathcal{P}(X) := \{A | A \subseteq X\}$, then a σ -algebra of subsets of X is $\mathcal{M} \subseteq \mathcal{P}(X)$, such that

1. $\emptyset \in \mathcal{M}$
2. If $E \in \mathcal{M}$, then $E^c := X \setminus E \in \mathcal{M}$
3. If $E_1, E_2, \dots \in \mathcal{M}$, then $\bigcup_{i=1}^\infty E_i \in \mathcal{M}$

Definition 1.28. If $\mathcal{M}$ is σ -algebra, we call $(X, \mathcal{M})$ a measurable space, and a set $E \in \mathcal{M}$ is called $\mathcal{M}$ -measurable.

Definition 1.29. For any set X and $a \in X$, we define $\delta_a : \mathcal{P}(X) \rightarrow [0, \infty]$ by $\delta_a(A) := \begin{cases} 1, & a \in A \\ 0, & \text{otherwise} \end{cases}$ is the **point measure** or **Dirac-delta measure** of a . We may sometimes abuse the notation and write $\delta_a(x)dx := d\delta_a$.

Definition 1.30. A function $\mu : \mathcal{M} \rightarrow [0, \infty]$ is called a **positive measure** if it satisfies **countable additivity**. Namely, for any pairwise disjoint sets $E_1, E_2, \dots$ in $\mathcal{M}$, we have

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i).$$

We call $(X, \mathcal{M}, \mu)$ a **measure space**.

Remark. We can similarly define a complex or a signed measure to be a function $\mathcal{M} \rightarrow \mathbb{C}$ or $\mathcal{M} \rightarrow [-\infty, \infty]$ that satisfies countable additivity.

Definition 1.31. Let $(X, \mathcal{T})$ be a topological space. We define the **support** of a Borel measure $\mu : \text{Bor}(X) \rightarrow [0, \infty]$ to be

$$\text{Supp}(\mu) := \{x \in X : \exists \text{ open } N \ni x, \text{ such that } \mu(N) = 0\}.$$

Definition 1.32. Let $M_+(\mathbb{R}^d)$ be the set of finite measures on $\mathbb{R}^d$, and $\mathcal{P}(\mathbb{R}^d) \subseteq M_+(\mathbb{R}^d)$ to be the set of probability measures. Also, let $M(\mathbb{R}^d) \supset M_+(\mathbb{R}^d)$ denote the set of finite signed measures.

Definition 1.33. Suppose ν is a signed measure, then $E \in \mathcal{M}$ is

1. **null** for ν if $\forall F \subseteq E, \nu(F) = 0$.
2. **positive** for ν if $\forall F \subseteq E, \nu(F) \geq 0$.
3. **negative** for ν if $\forall F \subseteq E, \nu(F) \leq 0$.

Theorem 1.14 (Hahn decomposition). *Let ν be a signed measure on $(X, \mathcal{M})$, there are $P, N \in \mathcal{M}$ such that $X = P \sqcup N$, and P is positive, N is negative. Moreover, this is unique in the sense that if $X = P' \sqcup N'$ is another such decomposition, then the symmetric difference $P \Delta P'$ is null.*

Definition 1.34. Let ν be a signed measure on $(X, \mathcal{M})$, and $P, N \in \mathcal{M}$ be as from Hahn decomposition, the **Jordan decomposition** of it is $\nu = \nu^+ - \nu^-$, where $\nu^+(E) := \nu(E \cap P), \nu^-(E) := -\nu(E \cap N)$ are positive measures.

Definition 1.35. Let ν be a signed measure on $(X, \mathcal{M})$, with Jordan decomposition $\nu = \nu^+ - \nu^-$, the **total variation** is

$$|\nu| := \nu^+ + \nu^-.$$

Definition 1.36. We also define $\|\mu\|_{M(\mathbb{R}^d)} := |\mu|(\mathbb{R}^d)$, where $|\mu|$ is the total variation of μ .

Remark. $M_+(\mathbb{R}^d)$ is not a $\mathbb{R}$ vector space, while $M(\mathbb{R}^d)$ is a Banach Space with $\|\cdot\|_{M(\mathbb{R}^d)}$.

Theorem 1.15 (Riesz-Representation for Measures). *Let X be a locally compact Hausdorff space. $(M(X), \|\cdot\|_{M(X)}) \cong C_0(X)^*$ isometrically, with $\mu \in M(X) \longleftrightarrow \Lambda \in C_0(X)^*$, such that*

$$\forall f \in C_0(X), \Lambda(f) = \int_{\mathbb{R}^d} f d\mu.$$

Moreover, $\|\Lambda\| = \|\mu\|_{M(X)} = |\mu|(X)$.

Remark. $\mathbb{R}^d$ is a locally compact Hausdorff space. Also, when X is compact, we have that $C_c(X) = C_0(X) = C_b(X) = C(X)$ by the extreme value theorem.

Remark. However, $M(\mathbb{R}^d) \subseteq C_b(\mathbb{R}^d)^*$, and the equality does not hold in general. This is analog to $(\ell^1)^* = \ell^\infty$, but $\ell^1 \subseteq (\ell^\infty)^*$.

Definition 1.37. Let $(X, \mathcal{M}, \mu)$ be a measure space, $(Y, \mathcal{M}_2)$ be a measurable space, and $M : X \rightarrow Y$ be a measurable function. The **push-forward** of μ by M is $M_{\#}\mu : \mathcal{M}_2 \rightarrow \mathbb{R}_+ \cup \{\infty\}$, defined by

$$M_{\#}\mu(E) := \mu(M^{-1}(E)).$$

Proposition 1.16. $(Y, \mathcal{M}_2, M_{\#}\mu)$ is a measure space.

Proposition 1.17. *For any measurable function $f : Y \rightarrow \mathbb{C}$ where the left-hand side makes sense, we have that*

$$\int_Y f dM_{\#}\mu = \int_X f \circ M d\mu.$$

1.7 Lower Semi-continuity

Definition 1.38. Let X be a vector space, and $F, G : X \rightarrow \mathbb{R} \cup \{\infty\}$. The **infimal convolution** of them is $F \square G : X \rightarrow \mathbb{R} \cup \{\infty\}$ defined by

$$(F \square G)(x) := \inf_{z \in X} F(x - z)G(z) = \inf_{a+b=x} F(a) + G(b).$$

Definition 1.39. Let X be a vector space, and $F : X \rightarrow \mathbb{R} \cup \{\infty\}$. We say F is **proper** if $\exists x \in X$ such that $F(x) < \infty$.

Proposition 1.18. Let X be a vector space, and $F, G : X \rightarrow \mathbb{R} \cup \{\infty\}$. If both F, G are bounded below/proper, we have that $F \square G$ is bounded below/proper as well.

Definition 1.40. Let X be a topological vector space. A function $F : X \rightarrow \mathbb{R} \cup \{\infty\}$ is **(sequentially) lower semi-continuous** if $\forall x \in X$ and any sequence $(x_n)_{n=1}^\infty$ that converges to x , we have $F(x) \leq \liminf_j F(x_j)$.

Proposition 1.19 (Identity). Let X be a vector space, and $F : X \rightarrow \mathbb{R} \cup \{\infty\}$. We have $\chi_{\{0\}} \square F = F$,

where $\chi_B(x) := \begin{cases} 0, & x \in B \\ \infty, & \text{otherwise} \end{cases}$ for $B \subseteq X, x \in X$.

Proposition 1.20 (Smoothing). Let $(X, \|\cdot\|)$ be a normed space, and $F : X \rightarrow \mathbb{R} \cup \{\infty\}$ is proper and bounded below. For any $\epsilon > 0$, we have $\frac{1}{\epsilon} \|\cdot\| \square F$ is $\frac{1}{\epsilon}$ -Lipschitz continuous.

Proposition 1.21 (Approximation). Let $(X, \|\cdot\|)$ be a normed space, and $F : X \rightarrow \mathbb{R} \cup \{\infty\}$ is proper, bounded below, and lower semi-continuous. For any $x \in X$, we have

$$F(x) = \lim_{\epsilon \searrow 0} \left(\frac{1}{\epsilon} \|\cdot\| \square F \right)(x)$$

monotonically.

Proof. Let $F_\epsilon := \frac{1}{\epsilon} \|\cdot\| \square F$. We notice that for any $x \in X$, $F_\epsilon(x) \leq F_\delta(x)$ for $\epsilon \geq \delta$. Thus, by monotone convergence theorem, we can define $a(x) := \lim_{\epsilon \searrow 0} F_\epsilon(x) \in \mathbb{R} \cup \{\infty\}$. Thus, for any $\epsilon > 0$, we can find $\delta \in (0, \epsilon)$, such that $|F_\delta(x) - a(x)| < \frac{1}{2}\epsilon$, and $y_\epsilon \in X$ such that $F(y_\epsilon) + \frac{1}{\delta}\|x - y_\epsilon\| < F_\delta(x) + \frac{1}{2}\epsilon$. We thus have

$$\begin{aligned} F(y_\epsilon) + \frac{1}{\epsilon}\|x - y_\epsilon\| &\leq \frac{1}{\delta}\|x - y_\epsilon\| \\ &< F_\delta(x) + \frac{1}{2}\epsilon \\ &< a(x) + \frac{1}{2}\epsilon + \frac{1}{2}\epsilon \\ &= a(x) + \epsilon. \end{aligned}$$

Since F is bounded below by some $c \in \mathbb{R}$, we have that

$$\begin{aligned} \|x - y_\epsilon\| &\leq \epsilon(a(x) - F(y_\epsilon)) + \epsilon^2 \\ &\leq \epsilon(a(x) - c) + \epsilon^2 \\ &\rightarrow 0 \end{aligned}$$

as $\epsilon \searrow 0$.

Thus, $y_\epsilon \rightarrow x$, and since F is lower semi-continuous, we have

$$\begin{aligned} F(x) &\leq \liminf_{\epsilon} F(y_\epsilon) \\ &\leq \liminf_{\epsilon} (a(x) + \epsilon - \frac{1}{\epsilon}\|x - y_\epsilon\|) \\ &\leq \liminf_{\epsilon} (a(x) + \epsilon) \\ &= a(x). \end{aligned}$$

On the other hand, for any $\epsilon > 0$, by choosing $0 + x = x$, we have $F_\epsilon(x) \leq \frac{1}{\epsilon}\|0\| + F(x) = F(x)$, so $a(x) \leq F(x)$.

This proves $F(x) = a(x) = \lim_{\epsilon \searrow 0} F_\epsilon(x) = \lim_{\epsilon \searrow 0} (\frac{1}{\epsilon}\|\cdot\| \square F)(x)$. $\square$

Proposition 1.22. *Let X be a topological vector space. Suppose $\{f_\lambda\}_{\lambda \in \Lambda}$ is a family of lower semi-continuous functions on X , then $f : X \rightarrow \mathbb{R} \cup \{\infty\}$ given by $f(x) := \sup_{\lambda \in \Lambda} f_\lambda(x)$ is lower semi-continuous.*

Proof. Given any $x \in X, \lambda \in \Lambda$, and any sequence $x_n \rightarrow x$. Since f_λ is lower semi-continuous, we have that

$$\begin{aligned} f_\lambda(x) &\leq \liminf_j f_\lambda(x_j) \\ &\leq \liminf_j \sup_{\lambda \in \Lambda} f_\lambda(x_j) \\ &= \liminf_j f(x_j). \end{aligned}$$

Since this holds for all $\lambda \in \Lambda$, we have that

$$\begin{aligned} f(x) &= \sup_{\lambda \in \Lambda} f_\lambda(x) \\ &\leq \liminf_j f(x_j). \end{aligned}$$

$\square$

Proposition 1.23. *Suppose $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{\infty\}$ is proper, lower semi-continuous and bounded below, then the functional $\Lambda(\mu) := \int_{\mathbb{R}^d} f d\mu$ is lower semi-continuous on $M_+(\mathbb{R}^d)$ with respect to narrow convergence.*

Proof. Consider $\tilde{f}_n := n\|\cdot\| \square f$, which is continuous by proposition 1.20. We can define $f_n := \min\{\tilde{f}_n, n\}$, which is also continuous. By proposition 1.21, we have that $f_n(x) \nearrow f(x)$ monotonically for each $x \in \mathbb{R}^d$. For each n , define $\Lambda_n(\mu) := \int_{\mathbb{R}^d} f_n d\mu$. Consider any sequence $(\mu_n)_{n=1}^\infty$ that converges narrowly to any μ in $M_+(\mathbb{R}^d)$. Since each f_n is continuous and bounded, we have that

$$\Lambda_n(\mu_j) = \int_{\mathbb{R}^d} f_n d\mu_j \rightarrow \int_{\mathbb{R}^d} f_n d\mu = \Lambda_n(\mu).$$

Thus Λ_n is continuous and thus lower semi-continuous with respect to the narrow convergence.

In addition, by Lebesgue's Monotone Convergence Theorem, we have for any $\mu \in M_+(\mathbb{R}^d)$,

$$\Lambda_n(\mu) = \int_{\mathbb{R}^d} f_n d\mu \nearrow \int_{\mathbb{R}^d} f d\mu = \Lambda(\mu).$$

Thus $\Lambda(\mu) = \sup_n \Lambda_n(\mu)$ is lower semi-continuous as well. $\square$

2 Monge and Kantorovich Problems

2.1 Monge Problem

Definition 2.1 (Monge problem). Consider the following setting: There is a pile of dirt and a hole with the same volume. We wish to move the pile of dirt into the hole at minimum cost.

Let X be the land space of the pile, $f : X \rightarrow \mathbb{R}_+$ be the height of the dirt, and the volume $V := \int_X f(x) dx < \infty$ is finite. Also, let Y be the land space of the hole, $g : Y \rightarrow \mathbb{R}_+$ be the depth of the hole, and its volume $\int_Y g(y) dy = V$. A **transport map** $M : X \rightarrow Y$ prescribes where the dirt at x should go.

M is **feasible** if for all open set $B \subseteq Y$, we have

$$\int_{M^{-1}(B)} f(x) dx = \int_B g(y) dy.$$

Namely, the dirt in the source of B exactly fills B .

For a cost function $c : X \times Y \rightarrow \mathbb{R}_+ \cup \{\infty\}$, then the cost of M is

$$\text{cost}(M) := \int_X c(x, M(x)) f(x) dx.$$

The problem is to find a feasible M that minimizes $\text{cost}(M)$.

Proposition 2.1. *When $X = Y = \mathbb{R}^d$ and M is a diffeomorphism with positive determinant, i.e. M is a bijection and M, M^{-1} are both differentiable, by the change of variable formula, we have for all open $B \subseteq \mathbb{R}^d$,*

$$\int_B g(y) dy = \int_{M^{-1}(B)} g(M(x)) \det(M'(x)) dx.$$

In this case, the feasibility condition becomes $\int_{M^{-1}(B)} f(x) dx = \int_{M^{-1}(B)} g(M(x)) \det(M'(x)) dx$. Since the integrands are both positive, we have that it holds if and only if

$$\forall x \in \mathbb{R}^d, f(x) = g(M(x)) \det(M'(x)).$$

Now we want to extend this problem to the case where the source and sink are discrete points. Thus, instead of functions f, g , we consider measures μ, ν .

Definition 2.2 (Monge problem with measures). Let $(X, \mathcal{M}, \mu), (Y, \mathcal{N}, \nu)$ be measure spaces, with $\mu(X) = \nu(Y) = 1$. A **transport map** $M : X \rightarrow Y$ is **feasible** if M is measurable, and for all ν -measurable set $B \in \mathcal{N}$, we have

$$\mu(M^{-1}(B)) = \nu(B).$$

Namely, $\nu = M_{\#}\mu$. We also define the cost function for M to be

$$\text{cost}(M) := \int_X c(x, M(x)) d\mu.$$

We wish to find a feasible M that minimizes $\text{cost}(M)$.

Proposition 2.2. *If we take $\mu(A) := \frac{1}{V} \int_A f(x) dx$ and $\nu(B) := \frac{1}{V} \int_B g(y) dy$, the original Monge problem is a special case of the Monge problem with measures.*

Notice that some problems that should have solutions are infeasible for Monge.

Example 2.1.1. Let $X = Y = \mathbb{R}$, and $\mu = \delta_1, \nu = \frac{1}{2}\delta_2 + \frac{1}{2}\delta_3$, there is no feasible solution since M cannot map 1 to both 2, 3.

2.2 Kantorovich Problem

To deal with this issue, we want to relax the problem to allow mass splitting.

Definition 2.3. Let $(X, \mathcal{M}, \mu), (Y, \mathcal{N}, \nu)$ be measure spaces, with $\mu(X) = \nu(Y) = 1$. A **transport plan** is a measure γ on $X \times Y$, where $\gamma(A \times B)$ is the mass moved from A to B , where $A \in \mathcal{M}, B \in \mathcal{N}$ are measurable subsets of X, Y .

Definition 2.4. Let $P_X : X \times Y \rightarrow X, P_Y : X \times Y \rightarrow Y$ be the projection maps $P_X(x, y) := x, P_Y(x, y) := y$. The **marginal measure** of γ on X is $P_{X\#}\gamma$. Namely, for all $A \subseteq X$, we have $P_{X\#}\gamma(A) := \gamma(A \times Y)$. Similarly, we can define $P_{Y\#}\gamma$.

Definition 2.5 (Kantorovich Problem). Let $(X, \mathcal{M}, \mu), (Y, \mathcal{N}, \nu)$ be measure spaces, with $\mu(X) = \nu(Y) = 1$. A transport plan γ is **feasible** if $P_{X\#}\gamma = \mu, P_{Y\#}\gamma = \nu$. We call the set of feasible transport plan $\Pi(\mu, \nu)$. We want to find a feasible transport plan $\gamma \in \Pi(\mu, \nu)$ that minimizes

$$\text{cost}(\gamma) := \int_{X \times Y} c(x, y) d\gamma.$$

Proposition 2.3. *There is always a feasible transport plan, defined by*

$$(\mu \times \nu)(A \times B) := \mu(A)\nu(B).$$

Proposition 2.4. *The Kantorovich Problem is a relaxation of the Monge Problem. Namely, for each feasible transport map M for the Monge Problem, there is a feasible transport plan γ with the same cost.*

Proof. Suppose M is a feasible transport map. Define

$$d\gamma(x, y) := d\mu(x)d\delta_{M(x)}(y).$$

Consider any $A \in \mathcal{M}$, we have

$$\begin{aligned} P_{X\#}\gamma(A) &= \gamma(A \times Y) \\ &= \int_{A \times Y} d\gamma(x, y) \\ &= \int_{A \times Y} d\mu(x)d\delta_{M(x)}(y) \\ &= \int_A \int_Y d\delta_{M(x)}(y)d\mu(x) \\ &= \int_A \delta_{M(x)}(Y) d\mu(x) \\ &= \mu(A). \end{aligned}$$

Similarly, for any $B \in \mathcal{N}$, we have

$$\begin{aligned} P_{Y\#}\gamma(B) &= \gamma(X \times B) \\ &= \int_{X \times B} d\mu(x)d\delta_{M(x)}(y) \\ &= \int_X \int_B d\delta_{M(x)}(y)d\mu(x) \\ &= \int_X \chi_B(M(x))d\mu(x) \\ &= \int_{M^{-1}(B)} d\mu(x) \\ &= \mu(M^{-1}(B)) \\ &= \nu(B), \end{aligned}$$

where $\chi_B(x) := \begin{cases} 1, & y \in B \\ 0, & \text{otherwise} \end{cases}$ is the characteristic map of B . Rigorously, we define

$$\gamma := (\mathbb{I}, M)_{\#}\mu.$$

Indeed, for any $A \in \mathcal{M}, B \in \mathcal{N}$, we have that

$$\begin{aligned} (\mathbb{I}, M)(u) \in A \times B &\iff u \in A, M(u) \in B \\ &\iff u \in A \cap M^{-1}(B). \end{aligned}$$

Thus,

$$\begin{aligned}
\int_{A \times B} d\mu(x) d\delta_{M(x)}(y) &= \int_X \chi_A(x) \int_Y \chi_B(y) d\delta_{M(x)}(y) d\mu(x) \\
&= \int_X \chi_A(x) \chi_B(M(x)) d\mu \\
&= \int_X \int_X \chi_A(x) \chi_{M^{-1}(B)}(x) d\mu \\
&= \mu(A \cap M^{-1}(B)) \\
&= \mu((\mathbb{I}, M)^{-1}(A \times B)) \\
&= (\mathbb{I}, M)_\# \mu(A \times B) \\
&= \gamma(A \times B).
\end{aligned}$$

This shows that γ is well-defined and is feasible.

Now, for the objective, we have

$$\begin{aligned}
\int_{X \times Y} c(x, y) d\gamma &= \int_{X \times Y} c(x, y) d\mu(x) d\delta_{M(x)}(y) \\
&= \int_X c(x, y) \int_Y d\delta_{M(x)}(y) d\mu(x) \\
&= \int_X c(x, M(x)) d\mu(x).
\end{aligned}$$

□

The optimal solutions may not be unique.

Example 2.2.1. Consider $X = Y = \mathbb{R}^2$, $c(x, y) = \|x - y\|^2$. Let $a = (0, 1)$, $b = (1, 0)$, $c = (0, -1)$, $d = (-1, 0)$, and $\mu = \frac{1}{2}(\delta_a + \delta_c)$, $\nu = \frac{1}{2}(\delta_b + \delta_d)$.

There are only two optimal solutions for Monge: $M(a) := b$, $M(c) := d$, or $M(a) := d$, $M(c) := b$.

There are infinitely many optimal solutions for Kantorovich: For any $\lambda \in [0, 1]$, we have that $\gamma := \frac{1}{2}\lambda(\delta_{(a,b)} + \delta_{(c,d)}) + \frac{1}{2}(1 - \lambda)(\delta_{(a,d)} + \delta_{(c,b)})$ is an optimal solution.

Example 2.2.2. Consider $X = Y = \mathbb{R}^2$, $c(x, y) = \|x - y\|^2$,

$$\mu(A) := \lambda(P_{y-axis}(A \cap (\{0\} \times [0, 1]))) = \lambda(\{y \in [0, 1] : (0, y) \in A\}),$$

and

$$\nu(B) := \frac{1}{2}\lambda(P_{y-axis}(B \cap (\{-1\} \times [0, 1]))) + \frac{1}{2}\lambda(P_{y-axis}(B \cap (\{1\} \times [0, 1]))),$$

where λ is the Lebesgue measure on $\mathbb{R}$.

We first consider the Monge problem. Notice that if M is feasible, we must have $\|x - M(x)\|^2 \geq 1$ for a.e. $x \in \{0\} \times [0, 1]$, since $M(x) \in \{\pm 1\} \times [0, 1]$. Thus,

$$cost(M) = \int_X \|x - M(x)\|^2 dx = \int_{\{0\} \times [0, 1]} \|x - M(x)\|^2 dx \geq 1.$$

However, we claim that 1 cannot be attained. Indeed, suppose for contradiction that

$$cost(M) = \int_{\{0\} \times [0, 1]} \|x - M(x)\|^2 dx = 1.$$

We must $\|x - M(x)\| = 1$ a.e. on $\{0\} \times [0, 1]$, which means $M(x)$ and x must have the same y -coordinates a.e., since $M(x) \in \{\pm 1\} \times [0, 1]$ a.e.. Now let

$$\Omega^+ := \{y \in [0, 1] : M(0, y) = (1, y)\}, \quad \Omega^- := \{y \in [0, 1] : M(0, y) = (-1, y)\}.$$

$$\begin{aligned}
M_{\#}\mu(B) &= \mu(M^{-1}(B)) \\
&= \lambda(P_{y\text{-axis}}(M^{-1}(B) \cap (\{0\} \times [0, 1]))) \\
&= \lambda(P_{y\text{-axis}}(M^{-1}(B) \cap (\{0\} \times (\Omega^+ \sqcup \Omega^-)))) \\
&= \lambda(P_{y\text{-axis}}(M^{-1}(B) \cap \{0\} \times \Omega^+)) + \lambda(P_{y\text{-axis}}(M^{-1}(B) \cap \{0\} \times \Omega^-)) \\
&= \lambda(P_{y\text{-axis}}(B \cap M(\{0\} \times \Omega^+))) + \lambda(B \cap M(\{0\} \times \Omega^-)) \\
&= \lambda(P_{y\text{-axis}}(B \cap \{1\} \times \Omega^+)) + \lambda(B \cap \{-1\} \times \Omega^-),
\end{aligned}$$

which is a contradiction since it cannot be the same as

$$\nu(B) = \frac{1}{2}(\lambda(P_{y\text{-axis}}(B \cap (\{-1\} \times [0, 1]))) + \lambda(P_{y\text{-axis}}(B \cap (\{1\} \times [0, 1])))$$

for all measurable $B \subseteq \mathbb{R}^2$.

2.3 Translation Problem

Definition 2.6. Assume $a \in X$. Let μ be a measure on X ,. Let $\tau_a : X \rightarrow X$ be defined by $\tau_a(x) := x + a$, then the **translation** of μ by a is $\tau_{a\#}\mu$.

Definition 2.7. A function $\phi : X \rightarrow \mathbb{R}$ is **convex** if

$$\forall x, y \in U, \forall \lambda \in [0, 1], \phi(\lambda x + (1 - \lambda)y) \leq \lambda\phi(x) + (1 - \lambda)\phi(y).$$

Theorem 2.5 (Jensen's Inequality - Integral Form). *If ϕ is convex, and μ is a probability measure on X , for any measurable function $f : X \rightarrow \mathbb{R}$ we have*

$$\phi\left(\int_X f(x)d\mu\right) \leq \int_X \phi(f(x))d\mu.$$

Proposition 2.6. Assume $X = Y = \mathbb{R}^d$, and $a \in \mathbb{R}^d$. Let μ be a probability measure on $\mathbb{R}^d$, with a finite mean; i.e. $\int_X \|x\|d\mu < \infty$, and ν be the translation of μ by a . Suppose $c(x, y) = \phi(x - y)$ for some convex function $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$, then the optimal Monge map is $M^* = \tau_a$.

Proof. We first check feasibility. By definition, $\nu = \tau_{a\#}\mu = M^*_{\#}\mu$.

Now we check the objective:

$$\begin{aligned}
\text{cost}(M^*) &= \int_{\mathbb{R}^d} c(x, M^*(x))d\mu \\
&= \int_{\mathbb{R}^d} \phi(x - M^*(x))d\mu \\
&= \int_{\mathbb{R}^d} \phi(-a)d\mu \\
&= \phi(-a) \int_{\mathbb{R}^d} d\mu \\
&= \phi(-a).
\end{aligned}$$

Let M be any feasible transport map.

$$\begin{aligned}
\text{cost}(M) &= \int_{\mathbb{R}^d} c(x, M(x))d\mu \\
&= \int_{\mathbb{R}^d} \phi(x - M(x))d\mu \\
&\geq \phi\left(\int_{\mathbb{R}^d} x - M(x)d\mu\right) \\
&\geq \phi\left(\int_{\mathbb{R}^d} xd\mu - \int_{\mathbb{R}^d} M(x)d\mu\right).
\end{aligned}$$

Now let $y := M(x)$, we have $d\mu(x) = d\nu(y)$ by feasibility, so

$$\begin{aligned} \text{cost}(M) &\geq \phi\left(\int_{\mathbb{R}^d} x d\mu - \int_{\mathbb{R}^d} M(x) d\mu\right) \\ &= \phi\left(\int_{\mathbb{R}^d} x d\mu(x) - \int_{\mathbb{R}^d} y d\nu(y)\right). \end{aligned}$$

Now let $u := y - a$, then $d\nu(y) = d\mu(u)$ by definition of $\nu = \tau_a \# \mu$, so

$$\begin{aligned} \text{cost}(M) &\geq \phi\left(\int_{\mathbb{R}^d} x d\mu(x) - \int_{\mathbb{R}^d} y d\nu(y)\right) \\ &= \phi\left(\int_{\mathbb{R}^d} x d\mu(x) - \int_{\mathbb{R}^d} (u + a) d\mu(u)\right) \\ &= \phi\left(-\int_{\mathbb{R}^d} a d\mu(u)\right) \\ &= \phi(-a) \\ &= \text{cost}(M^*). \end{aligned}$$

Thus, M^* is optimal. □

Remark. When ϕ is strictly convex, the above inequality is strict unless $x - M(x)$ is constant almost everywhere, which makes $M^* = \tau_a$ the unique optimal solution.

Example 2.3.1. When ϕ is a norm, or $\phi(v) = v^T K v$ for any $K \in \mathbb{S}_+^d$, ϕ is always convex.

2.4 Multi-marginal Optimal Transport

Definition 2.8. Given N spaces $X_1, \dots, X_N$, a cost function $c : X_1 \times \dots \times X_N \rightarrow \mathbb{R}_+$, and N probability measures μ_i on X_i .

The multi-marginal Kantorovich problem aims to find a probability measure γ on $X_1 \times \dots \times X_N$, that minimizes the cost

$$\int_{X_1 \times \dots \times X_N} c(x_1, \dots, x_N) d\gamma(x_1, \dots, x_N),$$

subject to the feasibility condition

$$\forall i \in [N], P_{X_i \#} \gamma = \mu_i.$$

The set of feasible transport plan is $\Pi(\mu_1, \dots, \mu_N)$.

The multi-marginal Monge problem aims to find transport maps $M_2, \dots, M_N$, where each $M_i : X_1 \rightarrow X_i$, that minimizes the cost

$$\text{cost}(\gamma) := \int_{X_1} c(x_1, M_2(x_1), \dots, M_N(x_1)) d\mu_1(x_1),$$

subject to the feasibility condition

$$\forall i \in [N], M_i \# \mu_1 = \mu_i.$$

Example 2.4.1 (Particle-tracking). Consider N snapshots, each with m particles in $\mathbb{R}^n$. We want to find a way to match the particles in each snapshots in a sensible way. Namely, every particle occurs exactly once in similar location in N snapshots. The Monge problem solves this using the distance as the cost function.

Remark. In the case of multi-marginal, then feasible region of Kantorovich problem is much larger than the one for Monge problem, since its dimension is much higher.

2.5 Fully Discrete Optimal Transport

Definition 2.9. An optimal transport problem is **fully discrete** if both X, Y are supported on a finite number of points.

Example 2.5.1. When X is finite, it is supported on a finite number of points. If X is infinite, but μ is a sum of finitely many point measures, it is also supported on a finite number of points.

Example 2.5.2. Suppose $X = Y = \mathbb{R}^d$, with $\mu = \frac{1}{\ell}(\delta_{x_1} + \cdots + \delta_{x_\ell})$, $\nu = \frac{1}{\ell}(\delta_{y_1} + \cdots + \delta_{y_\ell})$.

The Monge problem aims to find a permutation $\sigma : [\ell] \rightarrow [\ell]$, such that $\sum_{i=1}^{\ell} c(x_i, y_{\sigma(i)})$ is minimized.

The Kantorovich problem aims to find $w_{ij} \geq 0$ where $\forall j \in [\ell]$, $\sum_{i=1}^{\ell} w_{ij} = \frac{1}{\ell}$, $\forall i \in [\ell]$, $\sum_{j=1}^{\ell} w_{ij} = \frac{1}{\ell}$, such that $\sum_{i=1}^{\ell} \sum_{j=1}^{\ell} c(x_i, y_j) w_{ij}$ is minimized. Indeed, fix any $i \in [\ell]$, we have

$$\begin{aligned} \frac{1}{\ell} &= \mu(\{x_i\}) \\ &= P_{X\#} \gamma(\{x_i\}) \\ &= \gamma(\{x_i\} \times Y) \\ &= \int_{\{x_i\} \times Y} w_{kj} d\delta_{x_k, y_j} \\ &= \sum_{j=1}^{\ell} w_{ij}. \end{aligned}$$

In this case, the feasible region is $\frac{1}{\ell}W$, for $W \in \mathbb{R}^{\ell \times \ell}$, where $w_{ij} \geq 0$ and $\forall j \in [\ell]$, $\sum_{i=1}^{\ell} w_{ij} = 1$, $\forall j \in [\ell]$, $\sum_{j=1}^{\ell} w_{ij} = 1$. The class of matrices is called **doubly stochastic**, denoted DS_{ℓ} .

Remark. In this case, the Monge problem has ℓ unknowns, but the Kantorovich problem has ℓ^2 unknowns.

Proposition 2.7. DS_{ℓ} , the space of doubly stochastic matrices, is a polyhedron. In addition, it is nonempty and compact.

Proof. The constraints $\sum_{i=1}^{\ell} w_{ij} = \frac{1}{\ell}$ can be rewritten as $\sum_{i=1}^{\ell} w_{ij} \leq \frac{1}{\ell}$ and $\sum_{i=1}^{\ell} -w_{ij} \leq -\frac{1}{\ell}$, so DS_{ℓ} is a polyhedron. It is non-empty, since $\mathbb{I}_{\ell} \in DS_{\ell}$. It is compact since $DS_{\ell} \subseteq [0, 1]^{\ell \times \ell}$. $\square$

Corollary 2.8. Suppose $X = Y = \mathbb{R}^d$, with $\mu = \frac{1}{\ell}(\delta_{x_1} + \cdots + \delta_{x_\ell})$, $\nu = \frac{1}{\ell}(\delta_{y_1} + \cdots + \delta_{y_\ell})$. The Kantorovich problem always have an optimizer at the vertex of DS_{ℓ} .

Theorem 2.9 (Birkhoff-Von Neumann). *The vertices of DS_{ℓ} are exactly the $\ell!$ permutation matrices, namely, the matrices with exactly one 1 per row and column, and with other entries 0.*

Proof. ($\Leftarrow$). Assume Π is an $\ell \times \ell$ permutation matrix. Clearly $\Pi \in DS_{\ell}$.

Define $\hat{c} := -\Pi$, then $\langle \hat{c}, \Pi \rangle = -\ell$. Consider any other $M' \in DS_{\ell}$, we have that $M'_{ij} \in [0, 1]$ in all positions $\hat{C}_{ij} \neq 0$, so $M'_{ij} \hat{C}_{ij} = -M'_{ij} \geq -1$. If all of them are 1, we have $M' = \Pi$. However, if $M'_{ij} < 1$ at any (i, j) , we will have $M'_{ij} \hat{C}_{ij} = -M'_{ij} > -1$, so $\langle \hat{c}, \Pi \rangle > -\ell$. Thus, Π is exposed and a vertex.

($\Rightarrow$). Assume $M \in DS_{\ell}$ is not a permutation matrix. We call an entry **fractional** if it lives in $(0, 1)$. M has least one fractional entry since it is not a permutation matrix. Now form a bipartite graph with nodes U, V of $|U| = |V| = \ell$, and there is an edge between $i \in U, j \in V$ if M_{ij} is fractional. Observe that for any $w \in U \cup V$, if $\deg(w) \neq 0$, we must have $\deg(w) \geq 2$, since it must have at least two fractional entries to sum to 1.

Thus the bipartite graph has a cycle c . Since the node of the cycle alternate between U, V , the cycle must be even length. Thus, there is $\epsilon > 0$ such that if we update the entries of M by adding $\epsilon, -\epsilon, \epsilon, -\epsilon, \dots, \epsilon, -\epsilon$ or $-\epsilon, \epsilon, -\epsilon, \epsilon, \dots, -\epsilon, \epsilon$, while still maintaining the updated matrix in DS_{ℓ} . Let M' be the updated matrix with adding $\epsilon, -\epsilon, \dots$, and M'' be the updated matrix with adding $-\epsilon, \epsilon, \dots$. We have $M = \frac{1}{2}(M' + M'')$, but $M \neq M', M''$. Thus, M is not a vertex. $\square$

Corollary 2.10. Suppose $X = Y = \mathbb{R}^d$, with $\mu = \frac{1}{\ell}(\delta_{x_1} + \cdots + \delta_{x_\ell})$, $\nu = \frac{1}{\ell}(\delta_{y_1} + \cdots + \delta_{y_\ell})$, the Kantorovich problem always have an optimizer at the vertex of DS_{ℓ} , which induces a solution to the Monge problem with the same objective value.

2.5.1 Multi-marginal Case

Now consider the case of N -multi-marginal: For $i \in [N]$, there are ℓ_i discrete points in X_i , and weights $b_{i,j} \geq 0$ for $j \in [\ell_i]$, satisfying $\sum_{j=1}^{\ell_i} b_{i,j} = 1$. Namely, the measure $\mu_i := \sum_{j=1}^{\ell_i} b_{i,j} \delta_{x_{i,j}}$. The cost function will be $c(x_{1,i_1}, \dots, x_{N,i_N})$ for all possible $i_j \in [\ell_i]$.

In general, there is no Monge problem. A case where Monge is feasible is when $\ell_1 = \dots = \ell_N = \ell$, and $w_{ij} = \frac{1}{\ell}$ for all $i \in [N], j \in [\ell]$.

In fact, the Monge problem with $N = 3$ and the above assumption is NP-hard. However, the optimal Kantorovich solution is very sparse.

Theorem 2.11. *Let S denote the feasible region of N -multi-marginal Kantorovich problem. Let $\hat{v}$ be an extreme point of S , then $\hat{v}$ has at most $\ell_1 + \dots + \ell_N - N + 1$ non zero entries.*

Proof. Let the entries of $\hat{v}$ be denoted $\hat{v}_{i_1, \dots, i_N}$, and let $K := \left\{ w \in \mathbb{R}^{\ell_1 \dots \ell_N} : w \geq 0, \sum_{i_1=1}^{\ell_1} \dots \sum_{i_N=1}^{\ell_N} w_{i_1, \dots, i_N} = 1 \right\}$.

The marginal constraints for X_1 are

$$\begin{aligned} \sum_{i_2=1}^{\ell_2} \dots \sum_{i_N=1}^{\ell_N} w_{1,i_2, \dots, i_N} &= b_{1,1} \\ &\vdots \\ \sum_{i_2=1}^{\ell_2} \dots \sum_{i_N=1}^{\ell_N} w_{1,i_2, \dots, i_N} &= b_{1,\ell_1} \end{aligned}$$

With respect to K , one of these may be dropped since the sum of all weights are 1. Thus, $S = K \cap H_1 \cap \dots \cap H_p$, where $P = \sum_{i=1}^N (\ell_i - 1)$.

By Dubbin's theorem 1.7, $\hat{w}$ is in the convex combination of $\ell_1 + \dots + \ell_N - N + 1$ extreme points of K . However, the extreme points are exactly the vectors with exactly one non-zero entries. $\square$

3 Duality

The primal problem for Kantorovich problem with $N = 2$ is to find $v_{ij} \geq 0$ that minimizes

$$\sum_{i=1}^{\ell_1} \sum_{j=1}^{\ell_2} c_{ij} v_{ij},$$

such that

$$\begin{aligned} \forall i \in [\ell_1], \quad \sum_{j=1}^{\ell_2} v_{ij} &= w_i \\ \forall j \in [\ell_2], \quad \sum_{i=1}^{\ell_1} v_{ij} &= w'_j. \end{aligned}$$

From liner programming, the dual problem is to find $w_i \geq 0, w'_j \geq 0$ that maximizes

$$\sum_{i=1}^{\ell_1} w_i p_i + \sum_{j=1}^{\ell_2} w'_j q_j,$$

such that

$$\begin{aligned}\sum_{j=1}^{\ell_1} w_i &= 1 \\ \sum_{j=1}^{\ell_1} w'_j &= 1 \\ p_i + q_j &\leq c_{ij}.\end{aligned}$$

Also, the above problem is feasible and bounded, so it has an optimizer.

Now we want to extend this to the general case.

3.1 Existence of the Primal Problem

Definition 3.1. Let $(\mu_n)_{n=1}^\infty$ be a sequence in $M(\mathbb{R}^d)$. It **converges narrowly** to $\mu \in M(\mathbb{R}^d)$ if $\forall f \in C_b(\mathbb{R}^d)$, $\int_{\mathbb{R}^d} f d\mu_n \rightarrow \int_{\mathbb{R}^d} f d\mu$.

Remark. $(\mu_n)_{n=1}^\infty$ weak-* converges to μ if $\forall f \in C_0(\mathbb{R}^d)$, $\int_{\mathbb{R}^d} f d\mu_n \rightarrow \int_{\mathbb{R}^d} f d\mu$. However, it is not good enough, since $\delta_{x_n} \rightarrow 0$ for $\|x_n\| \geq n$, but each δ_{x_n} has mass 1, while 0 has mass 0.

Definition 3.2. A set $K \subseteq M_+(\mathbb{R}^d)$ is **tight** if

$$\lim_{R \rightarrow \infty} \sup_{\mu \in K} \mu(\mathbb{R}^d \setminus \bar{B}(0, R)) = 0.$$

Namely, all measures in K go to 0 outside of $\bar{B}(0, R)$ when $R \rightarrow \infty$ uniformly.

Theorem 3.1 (Prokhorov). *If $K \subseteq M_+(\mathbb{R}^d)$ is bounded and tight, it is relatively sequentially compact with respect to narrow convergence. Namely, for any sequence $(\mu_n)_{n=1}^\infty \subseteq K$, there is a subsequence $(\mu_{n_i})_{i=1}^\infty$ that narrowly converges to some $\mu \in M_+(\mathbb{R}^d)$.*

Proof. Let $(\mu_n)_{n=1}^\infty \subseteq K$ be a sequence from K . Since K is bounded and $M(\mathbb{R}^d)$ is dual to $C_0(\mathbb{R}^d)$, which is separable, by corollary 1.12, there is a subsequence $(\mu_{n_i})_{i=1}^\infty$ that weakly converges to some $\mu \in M(\mathbb{R}^d)$. Namely, $\forall g \in C_0(\mathbb{R}^d)$, we have

$$\int_{\mathbb{R}^d} g d\mu_{n_i} \rightarrow \int_{\mathbb{R}^d} g d\mu.$$

Given any $f \in C_b(\mathbb{R}^d)$, and any $\epsilon > 0$. By tightness, we can find $R > 0$, such that $\mu(\mathbb{R}^d \setminus \bar{B}(0, R)) \leq \frac{\epsilon}{3\|f\|_{C(\mathbb{R}^d)}}$ and $\forall i \in \mathbb{N}$, $\mu_{n_i}(\mathbb{R}^d \setminus \bar{B}(0, R)) \leq \frac{\epsilon}{3\|f\|_{C(\mathbb{R}^d)}}$.

Now consider the function $\xi_R : \mathbb{R}^d \rightarrow \mathbb{R}$, where $\xi_R(x) := \begin{cases} 1, & \|x\| \leq R \\ R + 1 - \|x\|, & R < \|x\| < R + 1 \\ 0, & \|x\| \geq R + 1. \end{cases}$

Since ξ_R is continuous, and $\text{Supp}(\xi_R) = \bar{B}(0, R + 1)$ is compact, we have that $\xi_R f \in C_c(\mathbb{R}^d)$. Thus, there is $N \in \mathbb{N}_+$, such that for all $i \geq N$, we have $|\int_{\mathbb{R}^d} \xi_R f d\mu_{n_i} - \int_{\mathbb{R}^d} \xi_R f d\mu| < \frac{\epsilon}{3}$. we also notice that

$(f - \xi_R f)(x) = 0$ for all $x \in \bar{B}(0, R)$.

$$\begin{aligned}
& \left| \int_{\mathbb{R}^d} f d\mu_{n_i} - \int_{\mathbb{R}^d} f d\mu \right| \\
&= \left| \int_{\mathbb{R}^d} f d\mu_{n_i} - \int_{\mathbb{R}^d} \xi_R f d\mu_{n_i} + \int_{\mathbb{R}^d} \xi_R f d\mu_{n_i} - \int_{\mathbb{R}^d} \xi_R f d\mu + \int_{\mathbb{R}^d} \xi_R f d\mu - \int_{\mathbb{R}^d} f d\mu \right| \\
&\leq \left| \int_{\mathbb{R}^d} f d\mu_{n_i} - \int_{\mathbb{R}^d} \xi_R f d\mu_{n_i} \right| + \left| \int_{\mathbb{R}^d} \xi_R f d\mu_{n_i} - \int_{\mathbb{R}^d} \xi_R f d\mu \right| + \left| \int_{\mathbb{R}^d} \xi_R f d\mu - \int_{\mathbb{R}^d} f d\mu \right| \\
&\leq \int_{\mathbb{R}^d} |f - \xi_R f| d\mu_{n_i} + \left| \int_{\mathbb{R}^d} \xi_R f d\mu_{n_i} - \int_{\mathbb{R}^d} \xi_R f d\mu \right| + \int_{\mathbb{R}^d} |\xi_R f - f| d\mu \\
&= \int_{\mathbb{R}^d \setminus \bar{B}(0, R)} |f - \xi_R f| d\mu_{n_i} + \left| \int_{\mathbb{R}^d} \xi_R f d\mu_{n_i} - \int_{\mathbb{R}^d} \xi_R f d\mu \right| + \int_{\mathbb{R}^d \setminus \bar{B}(0, R)} |\xi_R f - f| d\mu \\
&\leq \|f - \xi_R f\|_{C(\mathbb{R}^d)} \mu_{n_i}(\mathbb{R}^d \setminus \bar{B}(0, R)) + \left| \int_{\mathbb{R}^d} \xi_R f d\mu_{n_i} - \int_{\mathbb{R}^d} \xi_R f d\mu \right| + \|f - \xi_R f\|_{C(\mathbb{R}^d)} \mu(\mathbb{R}^d \setminus \bar{B}(0, R)) \\
&< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} \\
&= \epsilon.
\end{aligned}$$

This proves that $(\mu_{n_i})_{i=1}^\infty$ converges narrowly to μ . $\square$

Theorem 3.2 (Existence of Minimizer). *Consider the multi-marginal Kantorovich Problem as in definition 2.8. Suppose $X_i = \mathbb{R}^{d_i}$ for some $d_i \in \mathbb{N}_+$ and the cost function $c : \mathbb{R}^{d_1} \times \dots \times \mathbb{R}^{d_N} \rightarrow \mathbb{R} \cup \{\infty\}$ is lower semi-continuous and bounded from below, then the minimizer is attained.*

Proof. Let $d := \sum_{i=1}^N d_i$.

Since the Kantorovich Problem is always feasible, there is a sequence of feasible transport plans $(\gamma_j)_{j=1}^\infty \subset \mathbb{R}^d \subset \Pi(\mu_1, \dots, \mu_N)$ such that $\text{cost}(\gamma_j) \searrow \inf_{\gamma \in \Pi(\mu_1, \dots, \mu_N)} \text{cost}(\gamma)$. Notice that they are bounded since they all have norm 1.

We now claim they are tight as well. Indeed, given any $\epsilon > 0$, for each $i \in [N]$, we can choose $r_i > 0$ such that $\mu_i(\mathbb{R}^{d_i} \setminus B(0, r_i)) < \frac{\epsilon}{N}$, since $\mu_i(\mathbb{R}^{d_i}) = 1 < \infty$. Let $r := \max_{i \in [N]} r_i > 0$. For any $(x^1, \dots, x^n) \in \mathbb{R}^d \setminus \bar{B}(0, r)$, we must have each $x_i \in \mathbb{R}^{d_i} \setminus B(0, r_i)$. Now for any j , since γ_j is feasible, we must have

$$\begin{aligned}
\gamma_j(\mathbb{R}^d \setminus \bar{B}(0, r)) &\leq \gamma_j \left(\bigcup_{i=1}^N \{(x^1, \dots, x^n) \in \mathbb{R}^d : x^i \in \mathbb{R}^{d_i} \setminus B(0, r_i)\} \right) \\
&\leq \sum_{i=1}^N \gamma_j(\{(x^1, \dots, x^n) \in \mathbb{R}^d : x^i \in \mathbb{R}^{d_i} \setminus B(0, r_i)\}) \\
&= \sum_{i=1}^N \gamma_j(\mathbb{R}^{d_1} \times \dots \times (\mathbb{R}^{d_i} \setminus B(0, r_i)) \times \dots \times \mathbb{R}^{d_N}) \\
&= \sum_{i=1}^N (P_i)_\# \gamma_j(\mathbb{R}^{d_i} \setminus B(0, r_i)) \\
&= \sum_{i=1}^N \mu_i(\mathbb{R}^{d_i} \setminus B(0, r_i)) \\
&\leq \sum_{i=1}^N \frac{\epsilon}{N} \\
&= \epsilon.
\end{aligned}$$

By the Prokhorov theorem 3.1, there is a subsequence $(\gamma_{j_n})_{n=1}^\infty$ that narrowly converges to some $\gamma \in M_+(\mathbb{R}^d)$. By proposition 1.23, we have that

$$\text{cost}(\gamma) \leq \liminf_n \text{cost}(\gamma_{j_n}) = \lim_{j \rightarrow \infty} \text{cost}(\gamma_j) = \inf_{\gamma' \in \Pi(\mu_1, \dots, \mu_N)} \text{cost}(\gamma').$$

It suffices to show that γ is feasible. Namely, we want to show $(P_i)_\# \gamma = \mu_i$. Indeed, for any $i \in [N]$, and any $\phi \in C_0(\mathbb{R}^{d_i})$, we can define $\Phi(x_1, \dots, x_N) := \phi(x_i)$. Notice that $\Phi \in C_b(\mathbb{R}^d)$ (not $C_0(\mathbb{R}^d)$!), so

$$\begin{aligned} \int_{\mathbb{R}^{d_i}} \phi d\mu_i &= \int_{\mathbb{R}^{d_i}} \phi d(P_i)_\# \gamma_{j_n} \\ &= \int_{\mathbb{R}^d} \Phi d\gamma_{j_n} \\ &\rightarrow \int_{\mathbb{R}^d} \Phi d\gamma \\ &= \int_{\mathbb{R}^{d_i}} \phi d(P_i)_\# \gamma. \end{aligned}$$

Since $M(\mathbb{R}^{d_i}) = (C_0(\mathbb{R}^{d_i}))^*$, we have $\mu_i = (P_i)_\# \gamma$ as required. $\square$

3.2 Duality

Lemma 3.3 (Separation). *Let $(X, \|\cdot\|)$ be a normed vector space, $A, B \subseteq X$ be disjoint convex sets. Suppose one of them has a non empty interior, then $\exists t \in \mathbb{R}, \phi \in X^*$, such that*

$$\forall x \in \bar{A}, y \in \bar{B}, \phi(x) \leq t \leq \phi(y).$$

Proof. Assume $A^\circ \neq \emptyset$. Since $A^\circ \subseteq A$ is also convex and disjoint from B , by the first Geometric Hahn-Banach separation theorem 1.13, there is $t \in \mathbb{R}, \phi \in X^*$, such that

$$\forall x \in A^\circ, y \in B, \phi(x) < t \leq \phi(y).$$

Since $\phi \in X^*$ is continuous, $\phi(\bar{A}^\circ) \subseteq \overline{\phi(A^\circ)}$, and $\phi(\bar{B}) \subseteq \overline{\phi(B)}$. We thus have

$$\forall x \in \bar{A}^\circ = \bar{A}, y \in \bar{B}, \phi(x) \leq t \leq \phi(y).$$

Assume $B^\circ \neq \emptyset$, we can apply the previous case, and take $\bar{t} := -t \in \mathbb{R}, \bar{\phi} := -\phi \in X^*$. $\square$

Remark. For general topological spaces, we only have $\bar{A}^\circ \subseteq \bar{A}$, where equality may not be attained.

Definition 3.3. Let $(X, \|\cdot\|)$ be a normed vector space, and $f : X \rightarrow \mathbb{R} \cup \{\infty\}$ be proper. Its **Legendre transform** is $f^* : X^* \rightarrow \mathbb{R} \cup \{\infty\}$ defined by

$$f^*(\phi) := \sup_{x \in X} (\phi(x) - f(x)) = \sup_{x \in X} (\langle \phi | x \rangle - f(x)).$$

Theorem 3.4 (Fenchel-Rockafellar Duality). *Let $(X, \|\cdot\|)$ be a normed vector space, and $F, G : X \rightarrow \mathbb{R} \cup \{\infty\}$ both convex. Suppose there is $x_0 \in X$, such that both F, G are finite at x_0 , and one of them is continuous at x_0 , then*

$$\inf_{x \in X} (F(x) + G(x)) = \max_{\phi \in X^*} (-F^*(\phi) - G^*(-\phi)).$$

Proof. We first rewrite the right hand side as

$$\begin{aligned} \sup_{\phi \in X^*} (-F^*(\phi) - G^*(-\phi)) &= \sup_{\phi \in X^*} \left(- \sup_{x \in X} (\langle \phi | x \rangle - F(x)) - \sup_{y \in X} (\langle -\phi | y \rangle - G(y)) \right) \\ &= \sup_{\phi \in X^*} \left(\inf_{x, y \in X} (F(x) + G(y) + \langle \phi | y - x \rangle) \right). \end{aligned}$$

Take $x = y$, we have

$$\begin{aligned} \sup_{\phi \in X^*} \left(\inf_{x, y \in X} (F(x) + G(y) + \langle \phi | y - x \rangle) \right) &\leq \sup_{\phi \in X^*} \left(\inf_{x \in X} (F(x) + G(x) + \langle \phi | 0 \rangle) \right) \\ &= \sup_{\phi \in X^*} \left(\inf_{x \in X} (F(x) + G(x)) \right) \\ &= \inf_{x \in X} (F(x) + G(x)). \end{aligned}$$

We thus have

$$\sup_{\phi \in X^*} \left(\inf_{x, y \in X} (F(x) + G(y) + \langle \phi | y - x \rangle) \right) \leq \inf_{x \in X} (F(x) + G(x)).$$

Let $m := \inf_{x \in X} (F(x) + G(x))$. Now we want to show $m \leq \sup_{\phi \in X^*} (\inf_{x, y \in X} F(x) + G(y) + \langle \phi | y - x \rangle)$. Suppose $m = -\infty$, then $m \leq \sup_{\phi \in X^*} (-F^*(\phi) - G^*(\phi))$. Now suppose $m > -\infty$. Also, $m < \infty$ by taking $x = x_0$.

Consider the sets

$$\begin{aligned} C_1 &:= \{(x, \lambda) \in X \times \mathbb{R} : \lambda > F(x)\}, \\ C_2 &:= \{(y, \mu) \in X \times \mathbb{R} : \mu < m - G(y)\}, \end{aligned}$$

which are the epigraphs of $F, G - m$, thus convex. Notice that if $(x, \lambda) \in C_1 \cap C_2$, then $F(x) < \lambda < m - G(x)$, which means $F(x) + G(x) < m$, a contradiction with the choice of m . Thus C_1, C_2 are disjoint.

WLOG, we assume G is continuous of x_0 . Now consider $\lambda := m - G(x_0) - 1 < m - G(x_0)$, we have that (z_0, λ) lives in the interior of C_2 by the continuity of G . By the separation lemma, there is $(\phi, \alpha) \in (X^* \times \mathbb{R}) \setminus \{(0, 0)\}$ and some $d \in \mathbb{R}$, such that

$$\forall (y, \lambda) \in \bar{C}_2, (x, \mu) \in \bar{C}_1, \langle (\phi, \alpha) | (y, \lambda) \rangle \leq t \leq \langle (\phi, \alpha) | (x, \mu) \rangle.$$

Now take $x = y = x_0$, we have

$$\begin{aligned} \langle \phi | x_0 \rangle + \alpha \lambda &\geq t \\ &\geq \langle \phi | x_0 \rangle + \alpha \mu \\ &\implies \\ \alpha \lambda &\geq \alpha \mu \end{aligned}$$

for any $\lambda \geq F(x_0), \mu \leq m - G(x_0)$. Since $F(x_0) + G(x_0) \geq m$, we have $\lambda \geq \mu$, which means $\alpha \geq 0$. Now suppose for contradiction that $\alpha = 0$, then $\phi \neq 0$. Thus $\langle \phi | x \rangle \geq \langle \phi | y \rangle$ for any x, y such that $F(x) < \infty, G(y) < \infty$. By continuity and finiteness of G at x_0 , there is $\epsilon > 0$, such that $G(x_0 + \epsilon \eta) < \infty$ for any $\eta \in X$ such that $\|\eta\| = 1$. Thus

$$\langle \phi | z_0 \rangle \geq \langle \phi | z_0 + \epsilon \eta \rangle.$$

This implies that $\epsilon \langle \phi | \eta \rangle \leq 0$ for any $\|\eta\| = 1$. However, since ϕ is not 0, there is some x such that $\langle \phi | x \rangle \neq 0$. Taking $\eta_1 := \frac{x}{\|x\|}, \eta_2 := -\frac{x}{\|x\|}$, we have a contradiction. Thus $\alpha > 0$.

Now take $\bar{\phi} := \frac{\phi}{\alpha}$, and $\bar{t} := \frac{t}{\alpha}$, we have

$$\forall (y, \lambda) \in \bar{C}_2, (x, \mu) \in \bar{C}_1, \langle (\bar{\phi}, 1) | (y, \lambda) \rangle \leq \bar{t} \leq \langle (\bar{\phi}, 1) | (x, \mu) \rangle.$$

Taking $\lambda := F(x), \mu := m - G(y)$, we have

$$\langle \bar{\phi} | y \rangle + m - G(y) \leq \langle \bar{\phi} | x \rangle + F(x)$$

for any x, y such that $F(x), G(y) < \infty$. Thus

$$F(x) + G(y) + \langle -\bar{\phi} | y - x \rangle \geq m$$

for all $x, y \in X$. This shows that $m \leq \sup_{\phi \in X^*} (\inf_{x, y \in X} F(x) + G(y) + \langle \phi | y - x \rangle)$. We now have that

$$\inf_{x \in X} (F(x) + G(x)) = \sup_{\phi \in X^*} \left(\inf_{x, y \in X} (F(x) + G(y) + \langle \phi | y - x \rangle) \right) = \sup_{\phi \in X^*} (-F^*(\phi) - G^*(-\phi)),$$

and the right hand side is attained by $\bar{\phi}$. □

Theorem 3.5 (First Kantorovich Duality). *Let $\mu_1, \dots, \mu_N$ be probability measures on $\mathbb{R}^{d_1}, \dots, \mathbb{R}^{d_N}$, such that $Z_i := \text{Supp}(\mu_i) \subseteq \mathbb{R}^{d_i}$ are all compact. Suppose $c : Z_1 \times \dots \times Z_N \rightarrow \mathbb{R}$ is continuous under the supremum norm, then*

$$\begin{aligned} & \min \left\{ \int_{Z_1 \times \dots \times Z_N} cd\gamma : \gamma \in M_+(Z_1 \times \dots \times Z_N) \text{ such that } \forall i \in [N], (P_i)_\# \gamma = \mu_i \right\} \\ &= \sup \left\{ \sum_{i=1}^N \int_{Z_i} \phi_i d\mu_i : \phi_i \in C(Z_i), \text{ such that } \phi_1 \oplus \dots \oplus \phi_N \leq c \right\}, \end{aligned}$$

where $\phi_1 \oplus \dots \oplus \phi_N : (x_1, \dots, x_N) \mapsto \sum_{i=1}^N \phi_i(x_i)$.

Proof. Define $F, G : C(Z_1 \times \dots \times Z_N) \rightarrow \mathbb{R} \cup \{\infty\}$ by

$$F(u) := \begin{cases} \sum_{i=1}^N \int_{Z_i} \phi_i d\mu_i, & \exists \phi_i \in C(Z_i), \text{ such that } \phi_1 \oplus \dots \oplus \phi_N = u \\ \infty, & \text{otherwise.} \end{cases}$$

Also, define

$$G(u) := \begin{cases} 0, & u \geq -c \\ \infty, & \text{otherwise.} \end{cases}$$

We notice that F, G are both convex. In addition, F, G are both finite and G is continuous at any constant function $u_0 := C$, where $C > \max_{z \in Z_1 \times \dots \times Z_N} c(z)$.

We now have

$$\begin{aligned} \inf_{u \in C(Z_1 \times \dots \times Z_N)} (F(u) + G(u)) &= \inf \left\{ \sum_{i=1}^N \int_{Z_i} \phi_i d\mu_i : \phi_i \in C(Z_i), \text{ such that } \phi_1 \oplus \dots \oplus \phi_N \geq -c \right\} \\ &= \inf \left\{ \sum_{i=1}^N \int_{Z_i} -\tilde{\phi}_i d\mu_i : \tilde{\phi}_i \in C(Z_i), \text{ such that } \tilde{\phi}_1 \oplus \dots \oplus \tilde{\phi}_N \leq c \right\}. \end{aligned}$$

Now F^*, G^* are functionals on measures supported on $Z_1 \times \dots \times Z_N$, by theorem 1.15. We have

$$\begin{aligned} G^*(-\gamma) &= \sup_{u \in C(Z_1 \times \dots \times Z_N)} \left(- \int_{Z_1 \times \dots \times Z_N} ud\gamma - G(u) \right) \\ &= \sup_{u \in C(Z_1 \times \dots \times Z_N) \text{ such that } u \geq -c} - \int_{Z_1 \times \dots \times Z_N} ud\gamma \\ &= \sup_{u \in C(Z_1 \times \dots \times Z_N) \text{ such that } \tilde{u} \leq c} \int_{Z_1 \times \dots \times Z_N} \tilde{u} d\gamma \\ &= \begin{cases} \int_{Z_1 \times \dots \times Z_N} cd\gamma, & \gamma \in M_+(Z_1 \times \dots \times Z_N) \\ \infty, & \text{otherwise.} \end{cases} \end{aligned}$$

The last step is because if γ is negative in some subset, we can always define a continuous function that is arbitrarily large in it. Similarly,

$$\begin{aligned} F^*(\gamma) &= \sup_{u \in C(Z_1 \times \dots \times Z_N)} \left(\int_{Z_1 \times \dots \times Z_N} ud\gamma - F(u) \right) \\ &= \sup_{\phi_i \in C(Z_i)} \left(\int_{Z_1 \times \dots \times Z_N} (\phi_1 \oplus \dots \oplus \phi_N) d\gamma - \sum_{i=1}^N \int_{Z_i} \phi_i d\mu_i \right) \\ &= \sup_{\phi_i \in C(Z_i)} \left(\sum_{i=1}^N \int_{Z_i} \phi_i d(P_i)_\# \gamma - \sum_{i=1}^N \int_{Z_i} \phi_i d\mu_i \right) \\ &= \begin{cases} 0, & \text{if } \forall i \in [N], (P_i)_\# \gamma = \mu_i \\ \infty, & \text{otherwise.} \end{cases} \end{aligned}$$

Thus by Fenchel-Rockafellar Duality, we have

$$\begin{aligned}
& \max \left\{ - \int_{Z_1 \times \dots \times Z_N} c d\gamma : \gamma \in M_+(Z_1 \times \dots \times Z_N) \text{ such that } \forall i \in [N], (P_i)_\# \gamma = \mu_i \right\} \\
&= \max_{\gamma \in M(Z_1 \times \dots \times Z_N)} (-F^*(\phi) - G^*(-\phi)) \\
&= \max_{\gamma \in C(Z_1 \times \dots \times Z_N)^*} (-F^*(\phi) - G^*(-\phi)) \\
&= \inf_{u \in C(Z_1 \times \dots \times Z_N)} (F(u) + G(u)) \\
&= \inf \left\{ \sum_{i=1}^N \int_{Z_i} -\tilde{\phi}_i d\mu_i : \tilde{\phi}_i \in C(Z_i), \text{ such that } \tilde{\phi}_1 \oplus \dots \oplus \tilde{\phi}_N \leq c \right\}.
\end{aligned}$$

Taking negatives on both sides completes the proof. $\square$

More generally, when we only assume c is lower semi-continuous and bounded below, and taking $X_i := \mathbb{R}^{d_i}$ as in the proof of existence, we have this dual problem:

$$\sup_{(\phi_1, \dots, \phi_N) \in \mathcal{A}(c)} \sum_{i=1}^N \int_{Z_i} \phi_i d\mu_i, \tag{1}$$

where

$$\mathcal{A}(c) := \{(\phi_1, \dots, \phi_N) : \forall i \in [N], \phi_i \in Y_i(X_i), \phi_1 \oplus \dots \oplus \phi_N \leq c \text{ } (\mu_1 \otimes \dots \otimes \mu_N)\text{-a.e.}\},$$

is called the **admissible set of Kantorovich potentials**, with any $C_0(X_i) \subseteq Y_i \subseteq L^1(X_i, \mu_i)$.

3.3 Complementary Slackness

We first consider the discrete case, with $N = 2$. Suppose $(x_{ij})_{i \in [\ell_1], j \in [\ell_2]}$ is optimal for the primal problem, and $(p_i)_{i \in [\ell_1]}, (q_j)_{j \in [\ell_2]}$ is optimal for the dual problem, then

$$x_{ij}(c_{ij} - p_i - q_j) = 0$$

for all $i \in [\ell_1], j \in [\ell_2]$. Conversely, if $(x_{ij})_{i \in [\ell_1], j \in [\ell_2]}$ and $(p_i)_{i \in [\ell_1]}, (q_j)_{j \in [\ell_2]}$ are all feasible, and the complementary slackness holds, then they are optimal.

More generally, we have the following result:

Theorem 3.6 (Complementary Slackness). *Let $N = 2$, and the condition of theorem 3.5 holds. Suppose γ is a feasible for the primal problem, and $\phi_1, \dots, \phi_N$ is feasible for the dual problem, then they are both optimal if and only if*

$$\int_{\mathbb{R}^{d_1} \times \mathbb{R}^{d_N}} \left(c(x_1, \dots, x_N) - \sum_{i=1}^N \phi(x_i) \right) d\gamma = 0.$$

Corollary 3.7. *Let the condition of theorem 3.5 holds. Suppose γ is optimal for the primal problem, then for any optimal solution ϕ_1, ϕ_2 for the dual problem, and γ -a.e. $(x, y) \in \text{Supp}(\gamma)$, where $\nabla_x c(x, y), \nabla \phi_1(x)$ exist, we have*

$$\nabla \phi_1(x) = \nabla_x c(x, y).$$

Proof. By complementary slackness, $\int_{\mathbb{R}^{d_1} \times \mathbb{R}^{d_N}} (c(x, y) - \phi_1(x) - \phi_2(y)) d\gamma = 0$, so for γ -a.e. $(x, y) \in \text{Supp}(\gamma)$, we have $c(x, y) - \phi_1(x) - \phi_2(y) = 0$.

Now define $q(\tilde{x}) := c(\tilde{x}, y) - \phi_1(\tilde{x}) - \phi_2(y)$. Notice that $q(\tilde{x}) \geq 0$ for all $\tilde{x}$ by the feasibility of the dual problem. Thus, $q(x) = 0$ is a minimizer of q . If q is differentiable at x , we must have $\nabla_{\tilde{x}} q(x) = 0$, which means $\nabla_x c(x, y) - \nabla \phi_1(x) = 0$. $\square$

Corollary 3.8. *Let the condition of theorem 3.5 holds. Suppose γ is optimal for the primal problem, and it is a Monge solution; namely, $d\gamma(x, y) = d\mu(x)d\delta_{M(x)}(y)$, then for any optimal solution ϕ_1, ϕ_2 for the dual problem, and μ -a.e. x where $\nabla_x c(x, y), \nabla \phi_1(x)$ exist, we have*

$$\nabla \phi_1(x) = \nabla_x c(x, y)|_{y=M(x)}.$$

3.4 Brenier Theorem

For $N = 2$, since μ, ν are non-negative measures, the objective of the dual problem increase when ϕ_1, ϕ_2 increase. Thus, if we fix ϕ_1 , we can optimize ϕ_2 by taking $\phi_2(y)$ to be the largest value such that $\phi_1(x) \leq c(x, y)$ for all y .

Definition 3.4. Let X, Y be metric spaces, $c : X \times Y \rightarrow \mathbb{R}$ be a continuous function, and $\phi : X \rightarrow \mathbb{R} \cup \{-\infty\}, \psi : Y \rightarrow \mathbb{R} \cup \{-\infty\}$.

1. The c -transform of ϕ is $\phi^c : Y \rightarrow \mathbb{R} \cup \{\infty\}$ defined by

$$\phi^c(y) := \inf_{x \in X} (c(x, y) - \phi(x)).$$

2. The $\bar{c}$ -transform of ψ is $\psi^{\bar{c}} : X \rightarrow \mathbb{R} \cup \{\infty\}$ defined by

$$\psi^{\bar{c}}(x) := \inf_{y \in Y} (c(x, y) - \psi(y)).$$

3. A pair of functions ϕ, ψ are called c -conjugate if $\psi = \phi^c, \phi = \psi^{\bar{c}}$.

We notice that if $\phi \oplus \psi \leq c$, we have $\phi \oplus \psi \leq \phi \oplus \phi^c \leq c$ and $\phi \oplus \psi \leq \psi^{\bar{c}} \oplus \psi \leq c$. Also, $\phi^{c\bar{c}} \geq \phi, \psi^{\bar{c}c} \geq \psi, \phi^{c\bar{c}c} = \phi^c, \psi^{\bar{c}c\bar{c}} = \psi^{\bar{c}}$. Namely, we can apply the transform for at most three times to achieve a pair of feasible c -conjugates.

Definition 3.5. Two convex functions f, g are call **convex conjugates** of each other if they are the Legendre transformation of each other. Namely,

$$f^* = g, g^* = f.$$

Theorem 3.9 (Brenier's). Let $c(x, y) := \frac{1}{2}\|x - y\|^2$ on $\mathbb{R}^d \times \mathbb{R}^d$, and $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$ be such that $d\mu = f dx$ for some non-negative measurable function; namely, μ is absolutely continuous with respect to the Lebesgue measure. If both $X_1 := \text{Supp}(\mu), Y_1 := \text{Supp}(\nu)$ are compact, there is a unique Kantorovich solution (up to $\text{Supp}(\mu) \times \text{Supp}(\nu)$). In addition,

1. γ is a Monge map. Namely, there is a (μ -unique on $\text{Supp}(\mu)$) Monge map $M : \mathbb{R}^d \rightarrow \mathbb{R}^d$, such that $d\gamma(x, y) = d\mu(x)d\delta_{M(x)}(y)$. Namely, $\gamma = (\mathbb{I}, M)_{\#}\mu$.
2. $M(x) = x - \nabla\phi_1(x)$, where ϕ_1, ϕ_2 are optimal solutions for the dual problem.
3. For any $x \in \mathbb{R}^d$, we have that $\phi_1(x) > -\infty$, and ϕ_1 is locally Lipschitz.
4. For $u(x) := \phi_1(x) - \frac{\|x\|^2}{2}, v(x) := \phi_2(x) - \frac{\|x\|^2}{2}$, we have that $-u, -v$ are convex, and are convex conjugates, and $M(x) = -\nabla u(x)$.

Proof. Consider any optimal solution γ for the primal problem, and ϕ_1, ϕ_2 for the dual problem. Notice that WLOG, they can be taken to satisfy that

$$\phi_1(x) = \inf_{y \in Y_1} \left(\frac{1}{2}\|x - y\|^2 - \phi_2(y) \right), \quad \phi_2(y) = \inf_{x \in X_1} \left(\frac{1}{2}\|x - y\|^2 - \phi_1(x) \right),$$

for $x \in X_1, y \in Y_1$. Namely, ϕ_1, ϕ_2 are c -conjugates on (X_1, Y_1) . Otherwise just apply the c -transform and $\bar{c}$ -transform on (X_1, Y_1) for at most three times.

Now redefine $\phi_2 := \infty$ outside of Y_2 , and

$$\begin{aligned} \tilde{\phi}_1(x) &:= \phi_2^{\bar{c}}(x) \\ &= \inf_{y \in \mathbb{R}^d} \left(\frac{1}{2}\|x - y\|^2 - \phi_2(y) \right) \\ &= \inf_{y \in Y_1} \left(\frac{1}{2}\|x - y\|^2 - \phi_2(y) \right) \\ &> -\infty. \end{aligned}$$

Here we take the $\bar{c}$ -transform over the entire $\mathbb{R}^d$, but since $\phi_2(y) = -\infty$ outside of Y_1 , we may consider the infimum only on $\text{Supp}(\nu)$, which makes $\tilde{\phi}_1$ still agree with the original ϕ_1 on $\text{Supp}(\mu)$. Also, let

$$\tilde{\phi}_2(y) := \tilde{\phi}_1^c = \inf_{x \in \mathbb{R}^d} \left(\frac{1}{2} \|x - y\|^2 - \tilde{\phi}_1(x) \right),$$

which is the c -transform of $\tilde{\phi}_1$ on the entire $\mathbb{R}^d$. Now

$$\tilde{\phi}_2 = \tilde{\phi}_1^c = \phi_2^{\bar{c}c} \geq \phi_2.$$

On the other hand, for any $y \in Y_1$, we also have that

$$\begin{aligned} \tilde{\phi}_2(y) &= \inf_{x \in \mathbb{R}^d} \left(\frac{1}{2} \|x - y\|^2 - \tilde{\phi}_1(x) \right) \\ &\leq \inf_{x \in X_1} \left(\frac{1}{2} \|x - y\|^2 - \tilde{\phi}_1(x) \right) \\ &= \tilde{\phi}_1(y). \end{aligned}$$

Thus $\tilde{\phi}_2 \equiv \phi_2$ on Y_1 . Also,

$$\tilde{\phi}_2^{\bar{c}} = \phi_2^{\bar{c}c\bar{c}} = \phi_2^{\bar{c}} = \tilde{\phi}_1.$$

We have now get $\tilde{\phi}_1, \tilde{\phi}_2$ that are c -conjugates on $(\mathbb{R}^d, \mathbb{R}^d)$, and also $\tilde{\phi}_1|_{X_1} \equiv \phi_1|_{X_1}, \tilde{\phi}_2|_{Y_1} = \phi_2|_{Y_1}$ are also c -conjugates on (X_1, Y_1) . WLOG, for the following proof, we will call them ϕ_1, ϕ_2 .

Now, ϕ_1 is locally Lipschitz since it is the infimum of a family of locally Lipschitz (quadratic) functions. By the Rademacher theorem, since μ is absolutely continuous with respect to the Lebesgue measure, and ϕ_1 is locally Lipschitz, it is differentiable a.e.. Since μ is absolutely continuous with respect to the Lebesgue measure, ϕ_1 is differentiable μ -a.e.. Let $X_0 \subseteq \mathbb{R}^d$ be the set where ϕ_1 is differentiable. We have that

$$\begin{aligned} \gamma((X_1 \setminus X_0) \times Y_1) &= \mu(X_1 \setminus X_0) \\ &= \mu(\text{Supp}(\mu) \setminus X_0) \\ &= 0. \end{aligned}$$

Since $\text{Supp}(\gamma) \subseteq \text{Supp}(\mu) \times \text{Supp}(\nu) = X_1 \times Y_1$, it suffices to consider $X_0 \times Y_1$. By complementary slackness, for γ -a.e. $(x, y) \in \text{Supp}(\gamma) \cap (X_0 \times Y_1)$, we have that $\nabla \phi_1(x) = \nabla_x \left(\frac{1}{2} \|x - y\|^2 \right) = x - y$, which means $y = x - \nabla \phi_1(x)$. Namely, for each x , there is only one $y = x - \nabla \phi_1(x)$. Define $M(x) := x - \nabla \phi_1(x)$ on X_0 , with its graph $\mathcal{G}(M) := \{(x, M(x)) : x \in X_0\}$. We have that $\gamma(\text{Supp}(\gamma) \setminus \mathcal{G}(M)) = 0$. Namely, $\gamma = \gamma|_{\mathcal{G}(M)}$. We thus have

$$\begin{aligned} \gamma &= (\mathbb{I}_{\mathcal{G}(M)})_{\#} \gamma \\ &= (\mathbb{I}_{X_0}, M)_{\#} (P_1)_{\#} \gamma \\ &= (\mathbb{I}_{X_0}, M)_{\#} \mu. \end{aligned}$$

Namely, $d\gamma(x, y) = d\mu(x) d\delta_{M(x)}(y)$. We have

$$\begin{aligned} \nu &= (P_2)_{\#} \gamma \\ &= (P_2)_{\#} (\mathbb{I}, M)_{\#} \mu \\ &= M_{\#} \mu. \end{aligned}$$

Thus M is indeed a feasible map. Also, it is μ -unique since $\mu(\text{Supp}(\mu) \setminus X_0) = 0$.

Now we want to show that $-u, -v$ are convex conjugates.

$$\begin{aligned} \phi_1(x) &:= \phi_2^{\bar{c}}(x) \\ &= \inf_{y \in \mathbb{R}^d} \left(\frac{\|x - y\|^2}{2} - \phi_2(y) \right) \\ &= \inf_{y \in \mathbb{R}^d} \left(\frac{\|x\|^2}{2} - \langle x, y \rangle + \frac{\|y\|^2}{2} - \phi_2(y) \right). \end{aligned}$$

We thus have

$$\begin{aligned}
-u(x) &= -\phi_1(x) + \frac{\|x\|^2}{2} \\
&= -\inf_{y \in \mathbb{R}^d} \left(\frac{\|x\|^2}{2} - \langle x, y \rangle + \frac{\|y\|^2}{2} - \phi_2(y) \right) + \frac{\|x\|^2}{2} \\
&= -\inf_{y \in \mathbb{R}^d} \left(-\langle x, y \rangle + \frac{\|y\|^2}{2} - \phi_2(y) \right) \\
&= \sup_{y \in \mathbb{R}^d} \left(\langle x, y \rangle - \frac{\|y\|^2}{2} + \phi_2(y) \right) \\
&= \sup_{y \in \mathbb{R}^d} (\langle x, y \rangle + v(y)) \\
&= (-v)^*(x).
\end{aligned}$$

A theorem states that the Legendre transform is always convex. $\square$

Notice that if $d = 1$, we have that M is the derivative of a convex function, which is monotone. Also, the Brenier theorem may be shown to be true without assuming μ, ν being compactly supported. We will call the unique Monge map the **Brenier map** when the above theorem holds.

Example 3.4.1. Consider $d = 1$, and $d\mu = \chi_{[-\frac{1}{2}, \frac{1}{2}]}dx$, and $\nu = \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1$. It is easy to see that

$$M(x) = \begin{cases} 1, & x > 0 \\ -1, & x < 0, \text{ which is } \frac{d}{dx}(|\cdot|), \text{ and is differentiable almost everywhere (but not everywhere!).} \\ \text{any,} & x = 0 \end{cases}$$

Definition 3.6. A function $g(x)$ such that $g(x) + \frac{k}{2}\|x\|^2$ is convex is called **k-weakly convex**.

Note that $-\phi_1$ from above theorem is 1-weakly convex.

Theorem 3.10. Assume $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$, with μ being absolutely continuous with respect to the Lebesgue measure. Also, suppose $M : \mathbb{R}^d \rightarrow \mathbb{R}^d$ satisfies $M_{\#}\mu = \nu$, and $M(x) = x - \nabla\phi_1(x)$ for some continuous, a.e.-differentiable 1-weakly convex function $-\phi_1 : \mathbb{R}^d \rightarrow \mathbb{R}$, then M is optimal for the Kantorovich problem with cost function $c(x, y) := \frac{1}{2}\|x - y\|^2$.

Proof. Take

$$\phi_2(y) := \phi_1^c(y) = \inf_{x \in \text{Supp}(\mu)} \left(\frac{1}{2}\|x - y\|^2 - \phi_1(x) \right) =: \inf_{x \in \text{Supp}(\mu)} f_y(x),$$

where

$$\begin{aligned}
f_y(x) &:= \frac{1}{2}\|x - y\|^2 - \phi_1(x) \\
&= \frac{1}{2}\|x\|^2 - \langle x, y \rangle + \frac{1}{2}\|y\|^2 - \phi_1(x) \\
&= \left(-\phi_1(x) + \frac{1}{2}\|x\|^2 \right) - \langle x, y \rangle + \frac{1}{2}\|y\|^2
\end{aligned}$$

is convex in x . Thus, when $\nabla f_y(x) = x - y - \nabla\phi_1(x) = M(x) - y = 0$, we must have that x is a minimizer

of $f_{M(x)}$. Namely, $\phi_2(M(x)) = f_{M(x)}(x) = \frac{1}{2}\|x - M(x)\|^2 - \phi_1(x)$. Now we have

$$\begin{aligned}
& \int_{\text{Supp}(\mu) \times \text{Supp}(\nu)} (c(x, y) - \phi_1(x) - \phi_2(y)) d\gamma \\
& \int_{\text{Supp}(\mu)} (c(x, M(x)) - \phi_1(x) - \phi_2(M(x))) d\mu \\
& = \int_{\text{Supp}(\mu)} \left(\frac{1}{2}\|x - M(x)\|^2 - \phi_1(x) - \left(\frac{1}{2}\|x - M(x)\|^2 - \phi_1(x) \right) \right) d\mu \\
& = 0.
\end{aligned}$$

By complementary slackness, γ is optimal, so is M . $\square$

Definition 3.7. We say $c : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ satisfies the **twist condition** if for all $x_0 \in \mathbb{R}^d$, $\nabla_x c(x_0, y)$ is injective with respect to y .

Theorem 3.11. *If c satisfies the twist condition, the optimal Kantorovich solution is an optimal Monge solution.*

Example 3.4.2. For $c(x, y) = \|x - y\|$, it does not satisfy the twist condition. Indeed, $\nabla_x c(x, y) = \frac{x-y}{\|x-y\|}$. For any x_0 , consider $y = x_0 + \lambda v$, we have that

$$\begin{aligned}
\nabla_x c(x_0, y) &= \frac{x_0 - x_0 - \lambda v}{\|x_0 - x_0 - \lambda v\|} \\
&= \frac{-\lambda v}{\|\lambda v\|} \\
&= \frac{-v}{\|v\|},
\end{aligned}$$

which is independent of λ .

4 Wasserstein Distance

4.1 Wasserstein Distance

Definition 4.1. For $p \in [1, \infty)$, given $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$, we define the **p-Wasserstein distance**

$$W_p(\mu, \nu) := \left(\min_{\gamma: (P_1)_\# \gamma = \mu, (P_2)_\# \gamma = \nu} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^p d\gamma \right)^{\frac{1}{p}}.$$

Theorem 4.1. *The Wasserstein distances are metrics on $\mathcal{P}(\mathbb{R}^d)$.*

Proof. It is clear that $w_p(\mu, \nu) \geq 0$. Suppose $w_p(\mu, \nu) = 0$, and let γ be the optimal transport plan from μ to ν . Let $D := \{(x, y) \in \mathbb{R}^d \times \mathbb{R}^d : x = y\}$. We must have $\gamma((\mathbb{R}^d \times \mathbb{R}^d) \setminus D) = 0$, since $\|x - y\|^p > 0$ outside of D . Thus $\text{Supp}(\gamma) \subseteq D$. For any measurable $A \subseteq \mathbb{R}^d$, we have that

$$\begin{aligned}
\mu(A) &= \gamma(A \times \mathbb{R}^d) \\
&= \gamma((A \times \mathbb{R}^d) \cap D) \\
&= \gamma((\mathbb{R}^d \times A) \cap D) \\
&= \gamma(\mathbb{R}^d \times A) \\
&= \nu(A).
\end{aligned}$$

Symmetry is obvious.

We will only prove triangle inequality for $p > 1$. Let M be an optimal Monge map from μ to ν , and M' be an optimal Monge map from ν to ν' . We have them since $\|x - y\|^p$ satisfies the twist condition for $p > 1$. Thus $M' \circ M$ is a Monge map from μ to ν' . Let $f(x) := x - M(x)$, $g(x) := M(x) - M'(M(x))$, we have

$$\begin{aligned}
w_p(\mu, \nu') &\leq \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - M'(M(x))\|^p d\mu \right)^{\frac{1}{p}} \\
&\leq \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} (\|x - M(x) + M(x) - M'(M(x))\|)^p d\mu \right)^{\frac{1}{p}} \\
&= \|f + g\|_{L^p(\mathbb{R}^d)} \\
&\leq \|f\|_{L^p(\mathbb{R}^d)} + \|g\|_{L^p(\mathbb{R}^d)} \\
&= \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - M(x)\|^p d\mu \right)^{\frac{1}{p}} + \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|M(x) - M'(M(x))\|^p d\mu \right)^{\frac{1}{p}} \\
&\leq \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - M(x)\|^p d\mu \right)^{\frac{1}{p}} + \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|y - M'(y)\|^p d\nu \right)^{\frac{1}{p}} \\
&= w_p(\mu, \nu) + w_p(\nu, \nu').
\end{aligned}$$

□

4.2 Geodesic

Definition 4.2. Given $\mu_0, \mu_1 \in \mathcal{P}(\mathbb{R}^d)$. A curve $G : [0, 1] \rightarrow \mathcal{P}(\mathbb{R}^d)$ is called a **constant speed** or **arclength parametrized geodesic** with respect to W_2 between μ_0 and μ_1 if

$$\forall 0 \leq s \leq t \leq 1, W_2(\mu_t, \mu_s) = |t - s| W_2(\mu_0, \mu_1).$$

Proposition 4.2. Given $\mu_0, \mu_1 \in \mathcal{P}(\mathbb{R}^d)$. Let G be a constant speed geodesic, then it is the shortest path from μ_0 to μ_1 in the following sense: For all $0 = t_0 \leq t_1 \leq \dots \leq t_n = 1$, and $\nu_1, \dots, \nu_{n-1} \in \mathcal{P}(\mathbb{R}^d)$, with $\nu_0 := \mu_0, \nu_n = \mu_1$, we have that

$$W_2(\mu_0, \mu_1) = \sum_{i=1}^n W_2(\mu_i, \mu_{i-1}) \leq \sum_{i=1}^n W_2(\nu_i, \nu_{i-1}).$$

Definition 4.3. Given $\mu_0, \mu_1 \in \mathcal{P}(\mathbb{R}^d)$, and any optimal transport map γ from μ_0 to μ_1 , we define the **McCann Interpolation** between them to be $\mu_t := (g_t)_\# \gamma$, where $g_t(x, y) := (1 - t)x + ty$ for $t \in [0, 1]$.

Theorem 4.3 (McCann Interpolation). [3, Theorem 4.2.1] Suppose $\mu_0, \mu_1 \in \mathcal{P}(\mathbb{R}^d)$ are compactly supported and absolutely continuous with respect to the Lebesgue measure. Let M be the optimal Brenier map in theorem 3.9 from μ_0 to μ_1 , with cost function $c(x, y) = \frac{1}{2} \|x - y\|^2$. For all $t \in [0, 1]$, let

$$M_t(x) := (1 - t)x + tM(x), \quad \mu_t := (M_t)_\# \mu_0.$$

We have that μ_t is the McCann Interpolation, and $t \mapsto \mu_t$ is the unique constant speed geodesic between μ_0, μ_1 .

Example 4.2.1. Consider $a, b \in \mathbb{R}^d$, we have that $W_2(\delta_a, \delta_b) = \|a - b\|$, and the interpolate is $\mu_t = \delta_{(1-t)a + tb}$.

5 Wasserstein Gradient Flow

Consider the solution of $\frac{d}{dt} x(t) = -\nabla \phi(x(t))$, which is called **gradient flow**. We have that $\frac{d}{dt} \phi(x(t)) = \langle \nabla \phi(x(t)), \frac{d}{dt} x \rangle = -\|\nabla \phi(x(t))\|^2 \leq 0$. Thus, ϕ decreases along gradient flow trajectories, and only stalls at $\nabla \phi(x^*) = 0$. We want to apply a similar technique to a function $F : \mathcal{P}(\mathbb{R}^d) \rightarrow \mathbb{R}$.

Definition 5.1. Let

$$P_{2ac}(\mathbb{R}^d) := \left\{ \mu \in \mathcal{P}(\mathbb{R}^d) : \exists \check{\mu} \in L^1(\mathbb{R}^d) \text{ such that } d\mu = \check{\mu}dx, \int_{\mathbb{R}^d} \|x\|^2 d\mu < \infty \right\}$$

be the set of absolutely continuous (with respect to the Lebesgue measure) probability measures with finite second moment.

Definition 5.2. Given $F : P_{2ac}(\mathbb{R}^d) \rightarrow \mathbb{R}$, we say $g : \mathbb{R}^d \rightarrow \mathbb{R}$ is **the first variation** of F at μ , denoted $\delta F(\mu)$ if for all $\chi \in M(\mathbb{R}^d)$ such that for all small enough $\epsilon > 0$, $\mu + \epsilon\chi \in P_{2ac}(\mathbb{R}^d)$, we have

$$\int_{\mathbb{R}^d} \delta F(\mu) d\chi = \lim_{\epsilon \searrow 0} \frac{F(\mu + \epsilon\chi) - F(\mu)}{\epsilon} \in \mathbb{R}.$$

Notice that if F has the first variation at μ , the measure χ will always satisfy $\chi(\mathbb{R}^d) = 0$ to preserve the measures as probability measures.

Definition 5.3. We define the **Wasserstein gradient** of F at μ to be

$$\nabla^W F(\mu) := \nabla \delta F(\mu).$$

Theorem 5.1 (Integration by Parts for Divergence). *Assume the functions below are smooth enough. For $F : \mathbb{R}^n \rightarrow \mathbb{R}$, with its divergence $\nabla \cdot F := \text{Tr}(\nabla F) = \sum_{i=1}^n \partial_i F$, and $a : \mathbb{R}^n \rightarrow \mathbb{R}$, we have the following product rule:*

$$\nabla \cdot (aF) = \langle \nabla a, F \rangle + a \nabla \cdot (F).$$

Suppose μ is absolutely continuous with respect to the Lebesgue measure, from divergence theorem, we have that

$$\begin{aligned} \int_{\partial\Omega} \langle aF, \eta \rangle ds &= \int_{\Omega} \nabla \cdot (aF) d\mu \\ &= \int_{\Omega} \langle \nabla a, F \rangle d\mu + \int_{\Omega} a \nabla \cdot F d\mu. \end{aligned}$$

When $\text{Supp}(aF) \subset\subset \Omega$, we have the integration by parts formula

$$\int_{\Omega} \langle \nabla a, F \rangle d\mu = - \int_{\Omega} a \nabla \cdot F d\mu.$$

Definition 5.4. Suppose $\mu : \mathbb{R}_+ \rightarrow P_{2ac}(\mathbb{R}^d); t \mapsto \mu_t$ is a family of measures with $\check{\mu} : \mathbb{R}^d \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that $d\mu_t = \check{\mu}(\cdot, t)dx$, and $\partial_t \check{\mu}(\cdot, t)$ exists. We write $\frac{\partial}{\partial t} \mu_t$ to be the measure induced by $\frac{\partial}{\partial t} \check{\mu}(\cdot, t)$.

We will from now on always assume the assumption in the above definition holds.

Definition 5.5. Suppose $\mu : \mathbb{R}_+ \rightarrow P_{2ac}(\mathbb{R}^d)$ is a family of measures, we say $v : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}^d$ is **tangent** to μ at time t if it satisfies the **continuity equation**

$$\frac{\partial}{\partial t} \check{\mu}(\cdot, t) = -\nabla \cdot (\check{\mu}_t v)(\cdot, t).$$

v is tangent to μ if it is tangent to μ at all time $t \in \mathbb{R}_+$.

Proposition 5.2. [1, Proposition 16.3] Suppose $v : \mathbb{R}^d \times \mathbb{R}_+ \rightarrow \mathbb{R}^d$ is sufficiently smooth, then v is tangent to μ if and only if for any $\phi \in C_c^\infty(\mathbb{R}^d)$,

$$\int_{\mathbb{R}^d} \phi(x) d\left(\frac{\partial}{\partial t} \mu_t\right) = \int_{\mathbb{R}^d} \langle \nabla \phi(x), v(x, t) \rangle d\mu_t.$$

Proof. $\check{\mu}, v$ satisfies $\frac{\partial}{\partial t}\check{\mu}_t = -\nabla \cdot (\check{\mu}_t v)(\cdot, t)$ if and only if

$$\begin{aligned} \int_{\mathbb{R}^d} \frac{\partial}{\partial t} \check{\mu}(x, t) \phi(x) dx &= - \int_{\mathbb{R}^d} \nabla \cdot (\check{\mu}(x, t) v(x, t)) \phi(x) dx \\ \int_{\mathbb{R}^d} \phi(x) d\left(\frac{\partial}{\partial t} \mu_t\right) &= \int_{\mathbb{R}^d} \langle \nabla \phi(x), \check{\mu}(x, t) v(x, t) \rangle dx \\ &= \int_{\mathbb{R}^d} \langle \nabla \phi(x), v(x, t) \rangle \check{\mu}(x, t) dx \\ &= \int_{\mathbb{R}^d} \langle \nabla \phi(x), v(x, t) \rangle d\mu_t. \end{aligned}$$

□

Theorem 5.3. [2, Proposition 5.10] Given $F : P_{2ac}(\mathbb{R}^d) \rightarrow \mathbb{R}$. Suppose $\mu : \mathbb{R}_+ \rightarrow P_{2ac}(\mathbb{R}^d)$ is a family of measures such that F has a smooth first variation at μ_t for all t , then

$$\frac{d}{dt} F(\mu_t) = \langle \nabla^W F(\mu_t), v(\cdot, t) \rangle_{\mu_t} = \int_{\mathbb{R}^d} \langle \nabla^W F(\mu_t)(x), v(x, t) \rangle d\mu_t,$$

where $v : \mathbb{R}^d \times \mathbb{R}_+ \rightarrow \mathbb{R}^d$ is a tangent field.

Proof. One can show that $\delta F(\mu_t) \in C_c^\infty(\mathbb{R}^d)$, so

$$\begin{aligned} \int_{\mathbb{R}^d} \delta F(\mu_t) d\left(\frac{\partial}{\partial t} \mu_t\right) &= \int_{\mathbb{R}^d} \langle \nabla \delta F(\mu_t)(x), v(x, t) \rangle d\mu_t \\ &= \int_{\mathbb{R}^d} \langle \nabla^W F(\mu_t)(x), v(x, t) \rangle d\mu_t. \end{aligned}$$

On the other hand, taking $\chi := \frac{\partial}{\partial t} \mu_t$ in the definition of first variation, we have that

$$\begin{aligned} \int_{\mathbb{R}^d} \delta F(\mu_t) d\left(\frac{\partial}{\partial t} \mu_t\right) &= \lim_{\epsilon \searrow 0} \frac{F(\mu_t + \epsilon \frac{\partial}{\partial t} \mu_t) - F(\mu_t)}{\epsilon} \\ &= \lim_{\epsilon \searrow 0} \frac{F(\mu_t + \epsilon + o(\epsilon)) - F(\mu_t)}{\epsilon} \\ &= \frac{d}{dt} F(\mu_t). \end{aligned}$$

□

Definition 5.6. Given $F : P_{2ac}(\mathbb{R}^d) \rightarrow \mathbb{R}$. Suppose $\mu : \mathbb{R}_+ \rightarrow P_{2ac}(\mathbb{R}^d)$ is a family of measures such that F has a smooth first variation at μ_t for all t , we say μ_t is a **Wasserstein gradient flow** of F at t if μ_t is a solution to the PDE:

$$\frac{\partial}{\partial t} \check{\mu}(\cdot, t) = \nabla \cdot (\check{\mu}(\cdot, t) \nabla^W F(\mu_t)).$$

We say μ is a Wasserstein gradient flow of F if μ_t is a Wasserstein gradient flow of F at all t .

Corollary 5.4. Suppose μ_t is a Wasserstein gradient flow of F , we have that $v(\cdot, t) := -\nabla^W F(\mu_t)$ satisfies theorem 5.3,

$$\frac{d}{dt} F(\mu_t) = \langle \nabla^W F(\mu_t), -\nabla^W F(\mu_t) \rangle_{\mu_t} = -\|\nabla^W F(\mu_t)\|_{\mu_t}^2 \leq 0.$$

Thus, $F(\mu_t)$ is decreasing with t , and is stationary only if $\nabla^W F(\mu_t) = 0$.

Now, we would like to see that the Wasserstein gradient flow is actually a special tangent flow, which is naturally induced by Monge maps.

Theorem 5.5 (Cauchy-Lipschitz). [1, Theorem 16.2] Given $v : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}^d$ with some smoothness and boundedness conditions. There is an associated flow $X : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}^d$ such that $X(\bar{x}, \cdot)$ is a solution to the ODE

$$\begin{aligned} x(0) &= \bar{x} \\ \frac{d}{dt}x(t) &= v(x(t), t), \end{aligned}$$

and X satisfies the same smoothness and boundedness conditions.

Proposition 5.6. [1, Proposition 16.4] Given $v : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}^d$ with some smoothness and boundedness conditions. Suppose X is its associated flow as in theorem 5.5, then for any $\mu_0 \in P_{2ac}(\mathbb{R}^d)$,

$$\frac{\partial}{\partial t}\check{\mu}(\cdot, t) = -\nabla \cdot (\check{\mu}_t v)(\cdot, t)$$

for $\mu_t := X(\cdot, t)_{\#}\mu_0$. Namely, v is tangent to the family of measures μ generated by X .

Corollary 5.7. Let $\mu : \mathbb{R}_+ \rightarrow P_{2ac}(\mathbb{R}^d)$ be a family of measures, and $M_{t \rightarrow t+h}$ be the Brenier map from μ_t to μ_{t+h} given by theorem 3.9 under $c(x, y) = \frac{1}{2}\|x - y\|^2$. Suppose $v : \mathbb{R}^d \times \mathbb{R}_+ \rightarrow \mathbb{R}^d$, defined by

$$v(x, t) := \lim_{h \searrow 0} \frac{M_{t \rightarrow t+h}(x) - x}{h},$$

satisfies some smoothness and boundedness conditions, then v is tangent to μ .

Proof. One can show that the measures define by above proposition with $v(x, t) := \lim_{h \searrow 0} \frac{M_{t \rightarrow t+h}(x) - x}{h}$ is actually the original μ_t , thus $\frac{\partial}{\partial t}\check{\mu}_t = -\nabla \cdot (\check{\mu}_t v(\cdot, t))$. Indeed, $X(x, t)$ would just be where the Monge maps send x at time t . $\square$

We have shown that such a v defined above is indeed a tangent vector. Notice that in general, there may be more than one tangent vector that solves $\frac{\partial}{\partial t}\check{\mu} = -\nabla \cdot (\check{\mu}v)$.

Example 5.0.1. Let $d = 2$, and consider $\check{\mu}(x, t) := k(\|x\|^2) = k(x_1^2 + x_2^2)$ for some $k \in C^1(\mathbb{R}_+, \mathbb{R}_+)$. We have $\frac{\partial}{\partial t}\check{\mu} = 0$. The optimal Monge map is $M(x) = x$, so $v(x, t) = \lim_{h \searrow 0} \frac{M_{t \rightarrow t+h}(x) - x}{h} = 0$ is indeed tangent.

On the other hand, consider the rotational curve $v(x, t) := \alpha \begin{pmatrix} -x_2 \\ x_1 \end{pmatrix}$, we have

$$\begin{aligned} \nabla \cdot (\check{\mu}v) &= \alpha \nabla \cdot \left(\begin{pmatrix} -x_2 \\ x_1 \end{pmatrix} k(x_1^2 + x_2^2) \right) \\ &= \alpha (-x_2 k'(x_1^2 + x_2^2)(2x_1) + x_1 k'(x_1^2 + x_2^2)(2x_2)) \\ &= 0, \end{aligned}$$

which satisfies $\frac{\partial}{\partial t}\check{\mu} = -\nabla \cdot (\check{\mu}v)$ as well.

In fact, $v(x, t) := \lim_{h \searrow 0} \frac{M_{t \rightarrow t+h}(x) - x}{h}$ is the tangent vector that minimizes $\int_{\mathbb{R}^d} \|v(x, t)\|^2 d\mu_t$.

Theorem 5.8. Given $v : \mathbb{R}^d \times \mathbb{R}_+ \rightarrow \mathbb{R}^d$ with some smoothness and boundedness conditions. Let μ_t be the family of measures given by proposition 5.6. Suppose $v(\cdot, t) = -\nabla f(\cdot, t)$ for some $f : \mathbb{R}^d \times \mathbb{R}_+ \rightarrow \mathbb{R}$, then

$$v(x, t) = \lim_{h \searrow 0} \frac{M_{t \rightarrow t+h}(x) - x}{h}.$$

Proof. We give an intuition of the theorem for $t = 0$. We notice that for small enough $h > 0$, the associated flow $X(\cdot, h) \approx \mathbb{I} + hv(\cdot, 0)$ since $\frac{\partial}{\partial t}X = v$. Thus $\mu_h = X(\cdot, h)_{\#}\mu_0 \approx (\mathbb{I} + hv(\cdot, 0))_{\#}\mu_0 = (\mathbb{I} - h\nabla f(\cdot, 0))_{\#}\mu_0$. Also, one can show that $-hf$ is always 1-weakly convex for small enough h , so by theorem 3.10, $\mathbb{I} + hv(\cdot, 0)$ is (approximately) the optimal Monge map $M_{0 \rightarrow h}$, which shows $v(\cdot, 0) \approx \frac{M_{0 \rightarrow h} - \text{id}}{h}$. $\square$

Corollary 5.9. *Given $F : P_{2ac}(\mathbb{R}^d) \rightarrow \mathbb{R}$. Suppose $\mu : \mathbb{R}_+ \rightarrow P_{2ac}(\mathbb{R}^d)$ is a Wasserstein gradient flow of F , then*

$$\nabla^W F(\mu_t) = \lim_{h \searrow 0} \frac{M_{t \rightarrow t+h}(x) - x}{h}.$$

Proof. $v(\cdot, t) = -\nabla^W F(\mu_t) := \nabla(-\delta F(\mu_t))$ satisfies the above theorem, so $v(x, t) = \lim_{h \searrow 0} \frac{\tilde{M}_{t \rightarrow t+h}(x) - x}{h}$ for the induced family of measures $\tilde{\mu}$. It suffices to show $\mu = \tilde{\mu}$. Indeed, we have that v is tangent to $\tilde{\mu}$, so $\frac{\partial}{\partial t} \tilde{\mu}(\cdot, t) = -\nabla \cdot (\tilde{\mu}_t v)(\cdot, t)$. However, by definition of Wasserstein gradient flow, we also have $\frac{\partial}{\partial t} \mu(\cdot, t) = -\nabla \cdot (\mu_t v)(\cdot, t)$, so $\tilde{\mu} = \mu$. $\square$

6 Wasserstein Barycenter

Definition 6.1. Given $x_1, \dots, x_N \in \mathbb{R}^d$ and weights $\lambda_1, \dots, \lambda_N \geq 0$ such that $\sum_{i=1}^N \lambda_i = 1$, the **Euclidean weighted barycenter** is

$$B_\lambda(x_1, \dots, x_N) := \sum_{i=1}^N \lambda_i x_i.$$

Proposition 6.1. [3, Lemma 4.7] *Given $x_1, \dots, x_N \in \mathbb{R}^d$ and weights $\lambda_1, \dots, \lambda_N \geq 0$ such that $\sum_{i=1}^N \lambda_i = 1$, the Euclidean weighted barycenter is the unique minimizer of the Euclidean mean squared error*

$$f(y) := \sum_{i=1}^N \lambda_i \|y - x_i\|^2.$$

Definition 6.2. Given measures $\mu_1, \dots, \mu_N \in \mathcal{P}(\mathbb{R}^d)$, and weights $\lambda_1, \dots, \lambda_N \geq 0$ such that $\sum_{i=1}^N \lambda_i = 1$, the **Wasserstein Barycenter Problem** seeks to minimize

$$\int_{(\mathbb{R}^d)^N} c_\lambda(x_1, \dots, x_N) d\gamma \text{ such that } \forall i \in [N], (P_i)_\# = \mu_i,$$

where the cost function

$$c_\lambda(x_1, \dots, x_N) := \sum_{i=1}^N \lambda_i \|B_\lambda(x_1, \dots, x_N) - x_i\|^2.$$

Also, let γ be an optimal transport plan for the Wasserstein Barycenter Problem, we define the **Wasserstein Barycenter** of $\mu_1, \dots, \mu_N$ to be $\bar{\mu} := (B_\lambda)_\# \gamma$.

Example 6.0.1. Let $\mu_i := \delta_{x_i}$ for $x_1, \dots, x_N \in \mathbb{R}^d$, there is only one feasible transport plan $\delta_{x_1, \dots, x_N}$, so $(B_\lambda)_\# \lambda \delta_{x_1, \dots, x_N} = \delta_{B_\lambda(x_1, \dots, x_N)}$.

Proposition 6.2. *The cost function for the Wasserstein Barycenter Problem can be equivalently written as*

$$c_\lambda(x_1, \dots, x_N) = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \lambda_i \lambda_j \|x_i - x_j\|^2 = \sum_{i=1}^N \sum_{j=i+1}^N \lambda_i \lambda_j \|x_i - x_j\|^2.$$

Proof.

$$\begin{aligned}
& c_\lambda(x_1, \dots, x_N) \\
&:= \sum_{i=1}^N \lambda_i \|B_\lambda(x_1, \dots, x_N) - x_i\|^2 \\
&= \sum_{i=1}^N \lambda_i \left(\|B_\lambda(x_1, \dots, x_N)\|^2 - 2\langle B_\lambda(x_1, \dots, x_N), x_i \rangle + \|x_i\|^2 \right) \\
&= \|B_\lambda(x_1, \dots, x_N)\|^2 - 2 \sum_{i=1}^N \lambda_i \langle B_\lambda(x_1, \dots, x_N), x_i \rangle + \sum_{i=1}^N \lambda_i \|x_i\|^2 \\
&= \left\| \sum_{j=1}^N \lambda_j x_j \right\|^2 - 2 \sum_{i=1}^N \lambda_i \left\langle \sum_{j=1}^N \lambda_j x_j, x_i \right\rangle + \frac{1}{2} \left(\sum_{i=1}^N \lambda_i \|x_i\|^2 + \sum_{j=1}^N \lambda_j \|x_j\|^2 \right) \\
&= \sum_{j=1}^N \sum_{i=1}^N \lambda_j \lambda_i \langle x_j, x_i \rangle - 2 \sum_{i=1}^N \lambda_i \sum_{j=1}^N \lambda_j \langle x_j, x_i \rangle + \frac{1}{2} \left(\sum_{j=1}^N \lambda_j \sum_{i=1}^N \lambda_i \|x_i\|^2 + \sum_{i=1}^N \lambda_i \sum_{j=1}^N \lambda_j \|x_j\|^2 \right) \\
&= \sum_{i=1}^N \sum_{j=1}^N \lambda_i \lambda_j \left(\langle x_j, x_i \rangle - 2\langle x_j, x_i \rangle + \frac{1}{2}\|x_i\|^2 + \frac{1}{2}\|x_j\|^2 \right) \\
&= \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \lambda_i \lambda_j \|x_i - x_j\|^2 \\
&= \sum_{i=1}^N \sum_{j=i+1}^N \lambda_i \lambda_j \|x_i - x_j\|^2.
\end{aligned}$$

□

Corollary 6.3. [3, Lemma 4.8] Let $N = 2$, and μ_1, μ_2 be compactly supported absolutely continuous probability measures on $\mathbb{R}^d$, and $\lambda_1 = 1 - t, \lambda_2 = t$. The Wasserstein Barycenter is the McCann interpolation $(M_t)_\# \mu_0$.

Proof. We have $c_\lambda(x_1, x_2) = (t - 1)t\|x_1 - x_2\|^2$. Thus the optimal plan γ for the Wasserstein Barycenter Problem is just the Brenier map from μ_1 to μ_2 , which corresponds to the Monge map M . Now

$$\bar{\mu} = (B_\lambda)_\# \gamma = (B_\lambda)_\# (\mathbb{I}, M)_\# \mu_1 = ((1 - t)\mathbb{I} + tM)_\# \mu_1 = (M_t)_\# \mu_1.$$

□

Theorem 6.4 (Ganbo-Swiech). [3, Theorem 4.5.1] For any compactly supported and absolutely continuous measures $\mu_1, \dots, \mu_N$ on $\mathbb{R}^n$ and any weights $\lambda_1, \dots, \lambda_N \geq 0$ such that $\sum_{i=1}^N \lambda_i = 1$, the Kantorovich problem has a unique minimizer γ . In addition,

1. γ is a Monge map. Namely, there are (μ -unique on $\text{Supp}(\mu)$) Monge maps $M_i : \mathbb{R}^d \rightarrow \mathbb{R}^d$ for $i \in \{2, \dots, N\}$, such that

$$d\gamma(x_1, \dots, x_N) = d\mu(x_1) d\delta_{M_2(x_1)}(x_2) \cdots d\delta_{M_N(x_1)}(x_N).$$

Namely, $\gamma = (\mathbb{I}, M_2, \dots, M_N)_\# \mu$.

2. There are $\phi_1, \dots, \phi_N : \mathbb{R}^d \rightarrow \mathbb{R}$ that are optimal solutions for the dual problem, such that $u_i(x) := \frac{1}{\lambda_i} \phi_i(x) - \|x\|^2$ are strictly concave.
3. The gradients ∇u_i exists a.e., and are invertible, with $M_i(x) = (\nabla u_i)^{-1}(\nabla u_1(x_1))$.

Definition 6.3. Given measures $\mu_1, \dots, \mu_N, \mu \in \mathcal{P}(\mathbb{R}^d)$, and weights $\lambda_1, \dots, \lambda_N \geq 0$ such that $\sum_{i=1}^N \lambda_i = 1$. The **mean squared error**

$$MSE_\lambda(\mu) := \sum_{i=1}^N \lambda_i W_2(\mu_i, \mu)^2.$$

Theorem 6.5 (Argueh-Carlier). [3, Theorem 4.5.2] Given measures $\mu_1, \dots, \mu_N \in \mathcal{P}(\mathbb{R}^d)$, and weights $\lambda_1, \dots, \lambda_N \geq 0$ such that $\sum_{i=1}^N \lambda_i = 1$, we have that

$$\min_{\mu \in \mathcal{P}(\mathbb{R}^d)} MSE_\lambda(\mu) = \min_{\gamma \in \Pi(\mu_1, \dots, \mu_N)} \int_{\mathbb{R}^{Nd}} c_\lambda d\gamma.$$

In addition, the minimizer $\bar{\mu}$ of left-hand side is equal to $(B_\lambda)_\# \gamma$ for some optimizer of the right-hand side.

Proof. We will show $LHS \leq RHS$. Consider any optimizer of the right-hand side, let $\check{\mu} := (B_\lambda)_\# \gamma$ be a measure on $\mathbb{R}^d$. Let $\gamma_i := (B_\lambda, P_i)_\# \gamma$ be a measure on $\mathbb{R}^d \times \mathbb{R}^d$. It is easy to see that

$$(P_1)_\# \gamma_i = (B_\lambda)_\# \gamma = \check{\mu}, \quad (P_i)_\# \gamma_i = (P_i)_\# \gamma = \mu_i.$$

Thus γ_i is a feasible transport plan in $\Pi(\check{\mu}, \mu_i)$. We have that

$$\begin{aligned} \min_{\mu \in \mathcal{P}(\mathbb{R}^d)} MSE_\lambda(\mu) &\leq MSE_\lambda(\check{\mu}) \\ &= \sum_{i=1}^N \lambda_i W_2(\mu_i, \check{\mu})^2 \\ &\leq \sum_{i=1}^N \lambda_i \int_{\mathbb{R}^d \times \mathbb{R}^d} \|y - z\|^2 d\gamma_i \\ &= \sum_{i=1}^N \lambda_i \int_{\mathbb{R}^d \times \mathbb{R}^d} \|B_\lambda(x_1, \dots, x_N) - x_i\|^2 d\gamma \\ &= \int_{\mathbb{R}^d \times \mathbb{R}^d} c_\lambda d\gamma. \end{aligned}$$

□

Theorem 6.6 (Altschuler-Boix-Adsera). Let N be the number of marginals, each on $\mathbb{R}^d$, and each marginal has at most n deltas per marginal, then the Waaerstein Barycenter is computable in $(Nn)^d$ operations. However, the problem is NP-hard if d is allowed to vary.

7 Entropic Regularization and Sinkhorn

Definition 7.1. We consider the **entropy** function $q : \mathbb{R}_{++} \rightarrow \mathbb{R}$ by

$$q(t) := t \ln t.$$

Notice that this is the negation of traditional entropy $t \ln \left(\frac{1}{t}\right)$.

Proposition 7.1. Let q be defined as above, we have that $\lim_{t \rightarrow 0} q(t) = 0$, and q is strictly convex.

Proof. We have that $\lim_{t \rightarrow 0} q(t) = \lim_{s = -\ln t \rightarrow \infty} -e^{-s} s = 0$. Also, $q'(t) = \ln t + t \frac{1}{t} = \ln t + 1 > 0$, so q is strictly convex. □

Definition 7.2. We now extend $q : \mathbb{R} \rightarrow \mathbb{R} \cup \{\infty\}$ by

$$q(t) := \begin{cases} t \ln t & t > 0 \\ 0 & t = 0 \\ \infty & t < 0 \end{cases}.$$

Definition 7.3. Consider the fully discrete Kantorovich problem when $N = 2$. Namely,

$$c(x, y) = \sum_{i=1}^{\ell_1} \sum_{j=1}^{\ell_2} c_{ij} \delta_{a_i}(x) \delta_{b_j}(y), \quad \gamma = \sum_{i=1}^{\ell_1} \sum_{j=1}^{\ell_2} \gamma_{ij} \delta_{a_i} \otimes \delta_{b_j},$$

where each $(c_{ij}), (\gamma_{ij})$ is a $\ell_1 \times \ell_2$ matrix, and

$$\mu = \sum_{i=1}^{\ell_1} \mu_i \delta_{a_i}, \quad \nu = \sum_{j=1}^{\ell_2} \nu_j \delta_{b_j},$$

where $\mu \in \mathbb{R}^{\ell_1}, \nu \in \mathbb{R}^{\ell_2}$ such that $\sum_{i=1}^{\ell_1} \mu_i = \sum_{j=1}^{\ell_2} \nu_j = 1$. The problem is

$$\begin{aligned} \min \quad & \sum_{i=1}^{\ell_1} \sum_{j=1}^{\ell_2} c_{ij} \gamma_{ij} = \langle c, \gamma \rangle \\ \text{such that} \quad & \forall i \in [\ell_1], \sum_{j=1}^{\ell_2} \gamma_{ij} = \mu_i \\ & \forall j \in [\ell_2], \sum_{i=1}^{\ell_1} \gamma_{ij} = \nu_j \\ & \forall i \in [\ell_1], \forall j \in [\ell_2], \gamma_{ij} \geq 0. \end{aligned}$$

Assume that, WLOG, $\forall i \in [\ell_1], j \in [\ell_2], \mu_i > 0, \nu_j > 0$, the **entropic regularization** is the problem

$$\begin{aligned} \min \quad & \sum_{i=1}^{\ell_1} \sum_{j=1}^{\ell_2} (c_{ij} \gamma_{ij} + \epsilon \gamma_{ij} (\ln \gamma_{ij} - 1)) \\ \text{such that} \quad & \forall i \in [\ell_1], \sum_{j=1}^{\ell_2} \gamma_{ij} = \mu_i \\ & \forall j \in [\ell_2], \sum_{i=1}^{\ell_1} \gamma_{ij} = \nu_j \\ & \forall i \in [\ell_1], \forall j \in [\ell_2], \gamma_{ij} \geq 0, \end{aligned}$$

or, equivalently,

$$\begin{aligned} \min \quad & \sum_{i=1}^{\ell_1} \sum_{j=1}^{\ell_2} (c_{ij} \gamma_{ij} + \epsilon q(\gamma_{ij})) \\ \text{such that} \quad & \forall i \in [\ell_1], \sum_{j=1}^{\ell_2} \gamma_{ij} = \mu_i \\ & \forall j \in [\ell_2], \sum_{i=1}^{\ell_1} \gamma_{ij} = \nu_j, \end{aligned}$$

where $\epsilon > 0$ is called the **regularization parameter**.

Lemma 7.2. [3, Lemma 5.1] *The entropic regularization processes a unique minimizer γ . Moreover, each components $\gamma_{ij} > 0$.*

Proof. The domain of feasible γ is compact, and the objective is continuous on the domain. By the extreme value theorem, there is always a minimizer. The uniqueness is due to the strict convexity.

Let $\hat{\gamma}_{ij} := \mu_i \nu_j$. Suppose for contradiction that γ is an optimal solution, but $\gamma_{i'j'} = 0$. Let $\gamma_t := (1-t)\gamma + t\hat{\gamma}$ for $t \in [0, 1]$, we have that γ_t is feasible for all $t \in [0, 1]$, since $\gamma_0 = \gamma, \gamma_1 = \hat{\gamma}$, and the feasible region is convex. Now

$$\frac{d}{dt} \text{cost}(\gamma_t) = \sum_{i=1}^{\ell_1} \sum_{j=1}^{\ell_2} \left(c_{ij} \frac{d}{dt} \gamma_{t i' j'} + \epsilon \frac{d}{dt} q(\gamma_{t i' j'}) \right)$$

By assumption, for the $(i', j')^{th}$ entry, we have that

$$\begin{aligned} \frac{d}{dt} q(\gamma_{t i' j'}) &= \frac{d}{dt} q(t \hat{\gamma}_{i' j'}) \\ &= \ln(t \hat{\gamma}_{i' j'}) + 1. \end{aligned}$$

Thus, $\frac{d}{dt} \text{cost}(\gamma_t)|_{t=0} = -\infty$, which means the cost decreases as t increases from 0, which contradicts that $\gamma = \gamma_0$ is optimal. $\square$

Definition 7.4. We say a feasible solution x^* of an optimization problem

$$\begin{aligned} \min \quad & f(x) \\ \text{such that} \quad & \forall i \in [N], g_i(x) \leq 0 \end{aligned} ,$$

satisfies the **Karush–Kuhn–Tucker conditions** if there are Lagrange multipliers $\lambda_i \in \mathbb{R}_+$ for $i \in [N]$, such that both the stationary condition

$$\nabla f(x^*) + \sum_{i=1}^N \lambda_i \nabla g_i(x^*) = 0,$$

and the complementary slackness

$$\forall i \in [N], \lambda_i g_i(x^*) = 0$$

are satisfied.

Proposition 7.3. A solution x^* of an optimization problem

$$\begin{aligned} \min \quad & f(x) \\ \text{such that} \quad & \forall i \in [N], g_i(x) \leq 0 \quad , \\ & \forall j \in [M], h_j(x) = 0 \end{aligned}$$

satisfies the Karush–Kuhn–Tucker conditions if and only if there are Lagrange multipliers $\lambda_i \in \mathbb{R}_+$ for $i \in [N]$, and $\mu_j \in \mathbb{R}$ for $j \in [M]$, such that both the stationary condition

$$\nabla f(x^*) + \sum_{i=1}^N \lambda_i \nabla g_i(x^*) + \sum_{j=1}^M \mu_j \nabla h_j(x^*) = 0,$$

and the complementary slackness

$$\forall i \in [N], \lambda_i g_i(x^*) = 0$$

are satisfied for the inequality constraints.

Proof. We note that this is equivalent to a convex optimization problem of

$$\begin{aligned} \min \quad & f(x) \\ \text{such that} \quad & \forall i \in [N], g_i(x) \leq 0 \\ & \forall j \in [M], h_j(x) \leq 0 \\ & \forall j \in [M], -h_j(x) \leq 0 \end{aligned}$$

The stationary condition is there exists $\lambda_i \in \mathbb{R}_+$ for $i \in [N]$ and $\mu_j^+, \mu_j^- > 0$ for $j \in [M]$ such that

$$\nabla f(x^*) + \sum_{i=1}^N \lambda_i \nabla g_i(x^*) + \sum_{j=1}^M \mu_j^+ \nabla h_j(x^*) + \sum_{j=1}^M \mu_j^- \nabla (-h_j(x^*)) = 0$$

$$\nabla f(x^*) + \sum_{i=1}^N \lambda_i \nabla g_i(x^*) + \sum_{j=1}^M (\mu_j^+ - \mu_j^-) \nabla h_j(x^*) = 0.$$

Complementary slackness always holds for the equality constraints for any feasible solution. Thus taking $\mu_j := \mu_j^+ - \mu_j^- \in \mathbb{R}$ completes the proof. $\square$

Theorem 7.4 (Karush–Kuhn–Tucker). *For a differentiable convex objective with a polyhedron feasible region, a solution is optimal if and only if it satisfies the KKT conditions.*

Definition 7.5. Given $\epsilon > 0$ and the cost matrix $c \in \mathbb{R}^{\ell_1 \times \ell_2}$, the **Gibbs measure** $\bar{\gamma}^\epsilon \in \mathbb{R}^{\ell_1 \times \ell_2}$ is defined by

$$\bar{\gamma}_{ij}^\epsilon := \exp\left(-\frac{c_{ij}}{\epsilon}\right).$$

Theorem 7.5. [3, Theorem 5.2.1] *The optimizer γ of the entropic regularization is of the form $\gamma_{ij} = p_i \bar{\gamma}_{ij}^\epsilon q_j$ for some positive vectors $p \in \mathbb{R}_{++}^{\ell_1}, q \in \mathbb{R}_{++}^{\ell_2}$. Namely, $\gamma = \text{Diag}(p) \bar{\gamma}^\epsilon \text{Diag}(q)$.*

Proof. The derivative of the objective with respect to γ_{ij} is $c_{ij} + \epsilon q'(\gamma_{ij})$. The derivative of the i'^{th} row marginal condition is $\delta_{ii'}$. The derivative of the j'^{th} column marginal condition is $\delta_{jj'}$. Notice that γ is optimal if and only if it satisfies the KKT conditions. Since all the constraints $\gamma_{ij} \geq 0$ are not active and all the row and column marginal constraints are active, the complementary slackness is automatically satisfied for all the marginal constraints, and the Lagrange multipliers for the non-negativity constraints has to all be zero. Namely γ is optimal only if there are Lagrange multipliers $\lambda_i, \tilde{\lambda}_j \in \mathbb{R}$ (notice that each equality constraint is composed of two less than or equal to constraints of opposite signs, thus the difference between two positive Lagrange multipliers can be combined into any real number) for all $i \in [\ell_1], j \in [\ell_2]$ such that

$$\begin{aligned} c_{ij} + \epsilon q'(\gamma_{ij}) + \lambda_i + \tilde{\lambda}_j &= 0 \\ c_{ij} + \lambda_i + \tilde{\lambda}_j &= -\epsilon(\ln(\gamma_{ij}) + 1) \\ \ln(\gamma_{ij}) &= -\frac{c_{ij} + \lambda_i + \tilde{\lambda}_j}{\epsilon} - 1 \\ \gamma_{ij} &= \exp\left(-\frac{c_{ij}}{\epsilon}\right) \exp\left(-\frac{\lambda_i}{\epsilon}\right) \exp\left(-\frac{\tilde{\lambda}_j}{\epsilon} - 1\right) e^{-1} \\ &= \exp\left(-\frac{\lambda_i}{\epsilon}\right) \bar{\gamma}_{ij}^\epsilon \exp\left(-\frac{\tilde{\lambda}_j}{\epsilon} - 1\right). \end{aligned}$$

Now take $p_i := \exp\left(-\frac{\lambda_i}{\epsilon}\right) > 0, q_j := \exp\left(-\frac{\tilde{\lambda}_j}{\epsilon} - 1\right) > 0$. □

Since the KKT is necessary and sufficient, we have that the converse is also true.

Corollary 7.6. *Suppose $p \in \mathbb{R}_{++}^{\ell_1}, q \in \mathbb{R}_{++}^{\ell_2}$, and $\gamma = \text{Diag}(p) \bar{\gamma}^\epsilon \text{Diag}(q)$. If γ satisfies all the marginal constraints, γ is optimal for the entropic regularization.*

We observe that in this case, we have that for each $i \in [\ell_1]$,

$$\begin{aligned} \mu_i &= \sum_{j=1}^{\ell_2} \gamma_{ij} \\ &= \sum_{j=1}^{\ell_2} p_i \bar{\gamma}_{ij}^\epsilon q_j \\ &= p_i \sum_{j=1}^{\ell_2} \bar{\gamma}_{ij}^\epsilon q_j. \end{aligned}$$

Since all $\bar{\gamma}_{ij}^\epsilon, q_j$ are positive, we have

$$p_i = \frac{\mu_i}{\sum_{j=1}^{\ell_2} \bar{\gamma}_{ij}^\epsilon q_j}.$$

Similarly, for each $j \in [\ell_2]$,

$$q_j = \frac{\nu_j}{\sum_{i=1}^{\ell_1} \bar{\gamma}_{ij}^\epsilon p_i}.$$

This suggests the following algorithm:

Algorithm 1: Sinkhorn-Knopp

Initialize $q^0 \leftarrow e, p^0 \leftarrow e, k \leftarrow 0, \gamma^0 \leftarrow \text{Diag}(p^0)\bar{\gamma}^\epsilon \text{Diag}(q^0)$;

while γ^k is not optimal yet **do**

if k is even **then**

$$\begin{aligned} p_i^{k+1} &\leftarrow \frac{\mu_i}{\sum_{j=1}^{\ell_2} \bar{\gamma}_{ij}^\epsilon q_j^k}; \\ q^{k+1} &\leftarrow q^k; \end{aligned}$$

else

$$\begin{aligned} q_i^{k+1} &\leftarrow \frac{\nu_i}{\sum_{i=1}^{\ell_1} \bar{\gamma}_{ij}^\epsilon p_j^k}; \\ p^{k+1} &\leftarrow p^k; \end{aligned}$$

$$\gamma^{k+1} \leftarrow \text{Diag}(p^{k+1})\bar{\gamma}^\epsilon \text{Diag}(q^{k+1});$$

$$k \leftarrow k + 1$$

return γ^k

Write $x_i^k := \ln p_i^k, y_j^k := \ln q_j^k$, and we have $\gamma^k = \text{Diag}(\exp(x^k))\bar{\gamma}^\epsilon \text{Diag}(\exp(y^k))$. Notice that on even iterations,

$$\begin{aligned} p_i^{k+1} &= \frac{\mu_i}{\sum_{j=1}^{\ell_2} \bar{\gamma}_{ij}^\epsilon q_j^k} \\ &= \frac{\mu_i p_i^k}{\sum_{j=1}^{\ell_2} p_i^k \bar{\gamma}_{ij}^\epsilon q_j^k} \\ x_i^{k+1} &= \ln(\mu_i) + x_i^k - \ln \left(\sum_{j=1}^{\ell_2} \gamma_{ij}^k \right). \end{aligned}$$

Similarly, on odd iterations, we have

$$y_j^{k+1} = \ln(\nu_j) + y_j^k - \ln \left(\sum_{i=1}^{\ell_1} \gamma_{ij}^k \right).$$

Thus the Sinkhorn-Knopp algorithm can be equivalently written as:

Algorithm 2: Sinkhorn-Knopp

Initialize $x^0 \leftarrow 0, y^0 \leftarrow 0, k \leftarrow 0, \gamma^0 \leftarrow \text{Diag}(\exp(x^0))\bar{\gamma}^\epsilon \text{Diag}(\exp(y^0))$;

while γ^k is not optimal yet **do**

if k is even **then**

$$\begin{aligned} x_i^{k+1} &\leftarrow \ln(\mu_i) + x_i^k - \ln \left(\sum_{j=1}^{\ell_2} \gamma_{ij}^k \right); \\ y^{k+1} &\leftarrow y^k; \end{aligned}$$

else

$$\begin{aligned} y_j^{k+1} &\leftarrow \ln(\nu_j) + y_j^k - \ln \left(\sum_{i=1}^{\ell_1} \gamma_{ij}^k \right); \\ x^{k+1} &\leftarrow x^k; \end{aligned}$$

$$\gamma^{k+1} \leftarrow \text{Diag}(\exp(x^{k+1}))\bar{\gamma}^\epsilon \text{Diag}(\exp(y^{k+1}));$$

$$k \leftarrow k + 1$$

return γ^k

Remark. This algorithm is better at implementation since the updates are using addition after the natural log, which handles better with underflow.

Proposition 7.7. *Let*

$$f(x, y) := \sum_{i=1}^{\ell_1} \sum_{j=1}^{\ell_2} \bar{\gamma}_{ij}^\epsilon \exp(x_i + y_j) - \mu^T x - \nu^T y,$$

we have that f is convex. In addition, suppose (x^, y^*) is a minimizer of f , then $\gamma = \text{Diag}(\exp(x^*)) \bar{\gamma}^\epsilon \text{Diag}(\exp(y^*))$ is an optimizer of the entropic regularization.*

Proof. f is clearly convex.

Notice that

$$\frac{\partial f}{\partial x_i} = \sum_{j=1}^{\ell_2} \bar{\gamma}_{ij}^\epsilon \exp(x_i + y_j) - \mu_i,$$

so $\frac{\partial f}{\partial x_i} = 0$ if and only if

$$\mu_i = \sum_{j=1}^{\ell_2} \bar{\gamma}_{ij}^\epsilon \exp(x_i + y_j) = \sum_{j=1}^{\ell_2} \bar{\gamma}_{ij}^\epsilon p_i q_j,$$

namely, the i^{th} row marginal constraint is satisfied. Similar for the column marginal constraints. Since f is convex, we must have $\nabla_x f = \nabla_y f = 0$ at the minimizer (x^*, y^*) , so all the marginal constraints are satisfied. $\square$

Notice that the minimizer is unchanged if we add any constant α to all of c_{ij} . This scales $\bar{\gamma}^\epsilon = \exp(-c/\epsilon)$ by $\exp(-\alpha/\epsilon)$. WLOG, we can choose α such that $\sum_{i=1}^{\ell_1} \sum_{j=1}^{\ell_2} \bar{\gamma}_{ij}^\epsilon = 1$.

Corollary 7.8. *Suppose $\sum_{i=1}^{\ell_1} \sum_{j=1}^{\ell_2} \bar{\gamma}_{ij}^\epsilon = 1$, and for all $i \in [\ell_1], j \in [\ell_2]$, $\bar{\gamma}_{ij}^\epsilon \geq \delta > 0$, then $1 - f^* \leq \ln\left(\frac{1}{\delta}\right)$, where f^* is the minimum value of f .*

Proof. Let (x^*, y^*) be a minimizer of f . From above, we know $\mu_i = \sum_{j=1}^{\ell_2} \bar{\gamma}_{ij}^\epsilon \exp(x_i^* + y_j^*)$ for all $i \in [\ell_1]$. Now fix any $i \in [\ell_1], j \in [\ell_2]$,

$$\begin{aligned} \bar{\gamma}_{ij}^\epsilon \exp(x_i^* + y_j^*) &\leq \sum_{i'=1}^{\ell_1} \sum_{j'=1}^{\ell_2} \bar{\gamma}_{i'j'}^\epsilon \exp(x_i^* + y_j^*) \\ &= \sum_{i'=1}^{\ell_1} \mu_{i'} \\ &= 1. \end{aligned}$$

Thus $\exp(x_i^* + y_j^*) \leq \frac{1}{\bar{\gamma}_{ij}^\epsilon} \leq \frac{1}{\delta}$.

$$\begin{aligned} 1 - f^* &= 1 - \left(\sum_{i=1}^{\ell_1} \sum_{j=1}^{\ell_2} \bar{\gamma}_{ij}^\epsilon \exp(x_i + y_j) - \mu^T x^* - \nu^T y^* \right) \\ &= \mu^T x^* + \nu^T y^* \\ &\leq \ln\left(\frac{1}{\delta}\right), \end{aligned}$$

since each $x_i^* + y_j^* \leq \ln\left(\frac{1}{\delta}\right)$, and $\mu^T x^* + \nu^T y^*$ is an weighted combination of them. $\square$

Proposition 7.9. *In Sinkhorn-Knopp algorithm, for each odd step $k \geq 1$,*

$$\forall i \in [\ell_1], \sum_{j=1}^{\ell_2} \gamma_{ij}^k = \mu_i.$$

Also, for all even steps $k \geq 2$,

$$\forall j \in [\ell_2], \sum_{i=1}^{\ell_1} \gamma_{ij}^k = \nu_j.$$

Proof. Suppose k is odd, we have that for all $i \in [\ell_2]$,

$$\begin{aligned}
\sum_{j=1}^{\ell_2} \gamma_{ij}^k &= \sum_{j=1}^{\ell_2} p_i^k \bar{\gamma}_{ij}^\epsilon q_j^k \\
&= \sum_{j=1}^{\ell_2} \frac{\mu_i}{\sum_{j'=1}^{\ell_2} \bar{\gamma}_{ij'}^\epsilon q_{j'}^{k-1}} \bar{\gamma}_{ij}^\epsilon q_j^{k-1} \\
&= \frac{\mu_i}{\sum_{j'=1}^{\ell_2} \bar{\gamma}_{ij'}^\epsilon q_{j'}^{k-1}} \sum_{j=1}^{\ell_2} \bar{\gamma}_{ij}^\epsilon q_j^{k-1} \\
&= \mu_i.
\end{aligned}$$

Similarly, suppose k is even, we have that

$$\forall j \in [\ell_2], \sum_{i=1}^{\ell_1} \gamma_{ij}^{k+1} = \nu_j.$$

□

Corollary 7.10. *In Sinkhorn-Knopp algorithm, for each step $k \geq 0$,*

$$\sum_{i=1}^{\ell_1} \sum_{j=1}^{\ell_2} \gamma_{ij}^k = 1.$$

The case $k = 0$ is obvious since $\gamma^0 = \mathbb{I} \bar{\gamma}^\epsilon \mathbb{I} = \bar{\gamma}^\epsilon$.

Corollary 7.11. *In Sinkhorn-Knopp algorithm, for each step $k \geq 0$,*

$$f(x^k, y^k) = 1 - \mu^T x - \nu^T y.$$

Definition 7.6. The probability simplex $\Delta^\ell := \{x \in \mathbb{R}^\ell : x \geq 0, \sum_{i=1}^\ell x_i = 1\}$ is the set of all positive probabilities that contains ℓ discrete points.

Definition 7.7. Given $p, q \in \Delta^\ell$, the **Kullback-Leibler divergence** is

$$K(p||q) := \sum_{i=1}^\ell p_i \ln \left(\frac{p_i}{q_i} \right).$$

Remark. This coincides with the relative entropy $S(p||q)$.

Proposition 7.12. *Given any $p, q \in \Delta^\ell$, $K(p||q) \geq 0$ and the equality holds if and only if $p = q$.*

Proposition 7.13 (Pinsker Inequality). *Given any $p, q \in \Delta^\ell$,*

$$\|p - q\|_1^2 \leq 2K(p||q).$$

Lemma 7.14. *In Sinkhorn-Knopp algorithm, for each step $k \geq 1$, we have*

$$f(x^k, y^k) - f(x^{k+1}, y^{k+1}) = K(\mu||rs(\gamma^k)) + K(\nu||cs(\gamma^k)),$$

where the row sum and column sum vectors are $rs(\gamma^k) := \begin{pmatrix} \sum_{j=1}^{\ell_2} \gamma_{1j}^k \\ \vdots \\ \sum_{j=1}^{\ell_2} \gamma_{\ell_1 j}^k \end{pmatrix}$, $cs(\gamma^k) := \begin{pmatrix} \sum_{i=1}^{\ell_1} \gamma_{i1}^k \\ \vdots \\ \sum_{i=1}^{\ell_1} \gamma_{i\ell_2}^k \end{pmatrix}$.

Proof. Suppose k is even, we have that

$$\begin{aligned}
f(x^k, y^k) - f(x^{k+1}, y^{k+1}) &= (1 - 1) - \mu^T(x^k - x^{k+1}) - \nu^T(y^k - y^{k+1}) \\
&= \mu^T(x^{k+1} - x^k) \\
&= \sum_{i=1}^{\ell_1} \mu_i(x_i^{k+1} - x_i^k) \\
&= \sum_{i=1}^{\ell_1} \mu_i \ln \left(\frac{\mu_i}{\sum_{j=1}^{\ell_2} \gamma_{ij}^k} \right) \\
&= K(\mu || rs(\gamma^k)).
\end{aligned}$$

Also, $\nu = cs(\gamma^k)$, so $K(\nu || cs(\gamma^k)) = 0$.

Similarly, when k is odd, we have that

$$f(x^k, y^k) - f(x^{k+1}, y^{k+1}) = \nu^T(y^{k+1} - y^k) = K(\nu || cs(\gamma^k)),$$

and $K(\mu || rs(\gamma^k)) = 0$. □

Lemma 7.15. $f(x^1, y^1) \leq f(x^0, y^0)$.

Proof. Notice that

$$\begin{aligned}
\sum_{i=1}^{\ell_1} \sum_{j=1}^{\ell_2} \gamma_{ij}^1 \ln \left(\frac{\gamma_{ij}^1}{\gamma_{ij}^0} \right) &= \sum_{i=1}^{\ell_1} \sum_{j=1}^{\ell_2} \exp(x_1^1) \bar{\gamma}_{ij}^\epsilon \ln \left(\frac{\exp(x_1^1) \bar{\gamma}_{ij}^\epsilon}{\bar{\gamma}_{ij}^\epsilon} \right) \\
&= \sum_{i=1}^{\ell_1} \sum_{j=1}^{\ell_2} \exp(x_1^1) \bar{\gamma}_{ij}^\epsilon x_i^1 \\
&= \sum_{i=1}^{\ell_1} \sum_{j=1}^{\ell_2} \frac{\mu_i}{\sum_{j'=1}^{\ell_2} \bar{\gamma}_{ij'}^\epsilon} \bar{\gamma}_{ij}^\epsilon x_i^1 \\
&= \sum_{i=1}^{\ell_1} \mu_i x_i^1.
\end{aligned}$$

Similar to the analysis above,

$$\begin{aligned}
f(x^0, y^0) - f(x^1, y^1) &= \mu^T(x^1 - x^0) \\
&= \sum_{i=1}^{\ell_1} \mu_i x_i^1 \\
&= \sum_{i=1}^{\ell_1} \sum_{j=1}^{\ell_2} \gamma_{ij}^1 \ln \left(\frac{\gamma_{ij}^1}{\gamma_{ij}^0} \right) \\
&= K(\gamma^1 || \gamma^0) \\
&\geq 0.
\end{aligned}$$
□

Theorem 7.16. Suppose for all $i \in [\ell_1], j \in [\ell_2]$, $\bar{\gamma}_{ij}^\epsilon \geq \delta > 0$, we have that the Sinkhorn-Knopp algorithm has square convergence with respect to ℓ^1 error.

Proof. Except for the first step, on each iteration k , we have that $f(x^k, y^k)$ monotonically decreases by $K(\mu||rs(\gamma^k)) + K(\nu||cs(\gamma^k))$. On the other hand, $f(x^0, y^0) - f^* = 1 - f^* \leq \ln(\frac{1}{\delta})$. Suppose we have not terminated in $k + 1$ iterations, we must have that for some $j \in [k]$,

$$K(\mu||rs(\gamma^j)) + K(\nu||cs(\gamma^j)) \leq \frac{\ln(\frac{1}{\delta})}{k}.$$

Thus

$$\begin{aligned} (\|\mu - rs(\lambda^j)\|_1 + \|\nu - cs(\lambda^j)\|_1)^2 &\leq 2(\|\mu - rs(\lambda^j)\|_1^2 + \|\nu - cs(\lambda^j)\|_1^2) \\ &\leq 4(K(\mu||rs(\gamma^j)) + K(\nu||cs(\gamma^j))) \\ &\leq \frac{4 \ln(\frac{1}{\delta})}{k}. \end{aligned}$$

□

Remark. Notice that $\bar{\gamma}^\epsilon = \exp(\frac{-c}{\epsilon})$, so $\ln(\frac{1}{\delta}) \sim \frac{1}{\epsilon}$. Namely, larger ϵ results in faster convergence.

On the other hand, we note that the smaller ϵ is, the closer the entropic regularization problem is to the optimal transport problem.

Also, we note that the sparsity of the optimal transport solution is not preserved by Sinkhorn-Knopp algorithm.

Proposition 7.17. *Let $\gamma(\epsilon)$ be solution to the entropic regularization problems with parameter ϵ , then $\gamma^* := \lim_{\epsilon \searrow 0} \gamma(\epsilon)$ is the solution to original optimal transport problem such that among all optimal solutions, γ^* also minimizes $\sum_{i,j} q(\gamma_{ij})$.*

Proposition 7.18. *Given any $\eta > 0$, we can find some $\epsilon > 0$, such that $cost_{ER}[\gamma(\epsilon)] \leq cost_{ER}[\gamma^*] \leq cost[\gamma(\epsilon)] + \eta$.*

7.1 Computation for Separable Cost

One of the applications is to compute the Wasserstein distance between 2 grey-scale images, each of which can be viewed as a probability measure on $\ell_1 = \ell_2 = n_{row} \times n_{col}$ points. Each update of $p_i^{k+1} \leftarrow \frac{\mu_i}{\sum_{j=1}^{\ell_2} \bar{\gamma}_{ij}^\epsilon q_j^k}$ takes $O(\ell_2)$ operations, so in total $O(\ell_1 \ell_2) = O(n_{row}^2 n_{col})^2$ operations.

We note that this can be much faster if the cost is separable. Namely, $c((n_1, m_1), (n_2, m_2)) = c_1(n_1, n_2) + c_2(m_1, m_2)$ for $(n, m) \in [n_{row}] \times [n_{col}]$. For instance, $c((n_1, m_1), (n_2, m_2)) = (n_1 - n_2)^2 + (m_1 - m_2)^2$. In this case, let $s_{n_1, n_2} := \exp(-c_1(n_1, n_2)/\epsilon)$, and $t_{m_1, m_2} := \exp(-c_2(m_1, m_2)/\epsilon)$. We notice that in this case, to compute the update for $i = (n_1, m_1)$, we need

$$\begin{aligned} \sum_{j=1}^{\ell_2} \bar{\gamma}_{ij}^\epsilon q_j^k &= \sum_{n_2=1}^{n_{row}} \sum_{m_2=1}^{n_{col}} \exp\left(-\frac{c((n_1, m_1), (n_2, m_2))}{\epsilon}\right) q_{m_1, m_2}^k \\ &= \sum_{n_2=1}^{n_{row}} s_{n_1, n_2} \sum_{m_2=1}^{n_{col}} t_{m_1, m_2} q_{m_1, m_2}^k. \end{aligned}$$

By pre-computing $\sum_{m_2=1}^{n_{col}} t_{m_1, m_2} q_{m_1, m_2}^k$ once with n_{col} computations, we can reduce the update of each p_i to be only n_{row} computations. Thus in total $O(n_{col} + \ell_1 * n_{row}) = O(n_{row}^2 n_{col})$ operations in one iteration to update p . Similarly, in odd iterations, we only need $O(n_{row} n_{col}^2)$ operations.

8 Sliced Wasserstein Distance

Definition 8.1. Let $\theta \in S^d$. We defined $P_\theta : \mathbb{R}^d \rightarrow \mathbb{R}$ to be the projection onto the line through θ . Namely, $P_\theta(x) := x^T \theta$.

Definition 8.2. Given μ, ν on $\mathbb{R}^d$, we define the **sliced Wasserstein distance** between them to be

$$SW(\mu, \nu) := \int_{\theta \in S^d} W_2((P_\theta)_\# \mu, (P_\theta)_\# \nu) d\theta.$$

Theorem 8.1. Suppose $\mu = \sum_{i=1}^n \frac{1}{n} \delta_{a_i}, \nu = \sum_{j=1}^n \frac{1}{n} \delta_{b_j}$, then

$$W_2((P_\theta)_\# \mu, (P_\theta)_\# \nu) = \frac{1}{n} \sqrt{\sum_{i=1}^n (\tilde{a}_i^T \theta - \tilde{b}_i^T \theta)^2},$$

where $\tilde{a}_i$ and $\tilde{b}_i$ are reordered decreasingly by $P_\theta(a_i)$ and $P_\theta(b_i)$. Thus the computation only requires $O(n \log n)$ operations in sorting.

In fact, when the weights are not all $\frac{1}{n}$, we can still do this in $O(n \log n + nd)$. This works because the optimal transport in 1D is monotonic (derivative of a convex function).

Theorem 8.2. Let μ be a probability measure on $\mathbb{R}^d$. If we choose a random sequence $a_1, \dots, a_s$ according to μ , and define $\mu^s := \frac{1}{s} \sum_{i=1}^s \delta_{a_i}$, we have that $\mu^s \rightarrow \mu$ narrowly with high probability.

Theorem 8.3 (Stability). [3, Theorem 6.10.1] Given N compact subsets $X_i \subseteq \mathbb{R}^d$ for $i \in [N]$, and a Lipschitz continuous cost function $c : X_1 \times \dots \times X_N \rightarrow \mathbb{R}$, and N sequences of probability measures $\mu_i^s \rightarrow \mu_i$ narrowly as $s \rightarrow \infty$, we have

1. As $s \rightarrow \infty$, we have

$$\min_{\gamma \in \Pi(\mu_1^s, \dots, \mu_N^s)} \text{cost}(\gamma) \rightarrow \min_{\gamma \in \Pi(\mu_1, \dots, \mu_N)} \text{cost}(\gamma).$$

2. Every sequence of minimizers of the left hand side has a subsequence that converges to a minimizer of the right hand side.
3. There is a sequence of c -conjugate dual optimizers of the left hand side, normalized so that $\|\phi_i\|_\infty = 0$ for all $i \in [N]$, which has a subsequence converging to a dual optimizer of the right hand side.

Proof. We sketch the proof when $N = 2$.

Let (γ^s) be a sequence of minimizers of the left hand side. It is bounded (since probability measure has norm 1) and tight (since $X_1 \times X_2$ is compact). By the Prokhorov theorem 3.1, there is a subsequence (γ^{s_k}) that narrowly converged to some γ . Now for any $\phi \in C_b(X_1)$

$$\begin{aligned} \int_{X_1} \phi d\mu &= \lim_{s \rightarrow \infty} \int_{X_1} \phi d\mu^s \\ &= \lim_{k \rightarrow \infty} \int_{X_1} \phi d\mu^{s_k} \\ &= \lim_{k \rightarrow \infty} \int_{X_1} \Phi d\gamma^{s_k} \\ &= \int_{X_1} \Phi d\gamma \\ &= \int_{X_1} \phi d(P_1)_\# \gamma, \end{aligned}$$

since $\Phi(x_1, x_2) := \phi(x_1)$ is in $C_b(X_1 \times X_2)$. Thus $(P_1)_\# \gamma = \mu$. Similarly, $(P_2)_\# \gamma = \nu$. This shows that γ is feasible. Now take any sequence (ϕ_1^s, ϕ_2^s) of dual optimizers of the left hand side. Notice that the dual maps may have different supports. To solve this problem, we redefine $\tilde{\phi}_2^s(x_2) := \phi_1^{s^c}(x_2) = \inf_{x \in X_1} (c(x_1, x_2) - \phi_1(x_1))$ to be the c -transform over the entire X_1 . Since c is Lipschitz continuous, so is $\tilde{\phi}_2$, a infimum of Lipschitz continuous functions. Also, redefine $\tilde{\phi}_2 := \phi_2^c$.

One can show that $(\tilde{\phi}_1^{s_k}, \tilde{\phi}_2^{s_k})$ are convergent to some (ϕ_1, ϕ_2) that is dual feasible for the right hand side. Complementary slackness show the optimality. $\square$

9 Semi-discrete Optimal Transport

Suppose $\mu = \sum_{i=1}^n \mu_i \delta_{a_i}$ is discrete, then the dual problem is finding $\phi \in \mathbb{R}^n, \psi : \mathbb{R}^d \rightarrow \mathbb{R}$ that minimizes

$$\sum_{i=1}^n \mu_i \phi_i + \int_{\mathbb{R}^d} \psi d\nu,$$

such that $\forall i \in [n], y \in \mathbb{R}^d, \phi_i + \psi(y) \leq \|a_i - y\|^2$.

Notice that $\psi(y) = \min_{i \in [n]} \{\|y - a_i\|^2 - \phi_i\}$. This naturally partitions $\mathbb{R}^d$ to n parts.

Definition 9.1. Let $\phi \in \mathbb{R}^n$, and given $(a_i)_{i=1}^n \subset \mathbb{R}^n$, we define the i^{th} **Laguerre cells** to be

$$L_i(\phi) = \left\{ y \in \mathbb{R}^d : \forall j \in [n], \|y - a_i\|^2 - \phi_i \leq \|y - a_j\|^2 - \phi_j \right\}.$$

Note that L_i contains the points that are closest to the ball $B(a_i, \phi_i)$ than other balls.

Proposition 9.1. Let the Laguerre cells be defined as above, we have that $\bigcup_{i=1}^n L_i(\phi) = \mathbb{R}^d$, and $L_i(\phi)^o \cap L_j(\phi)^o = \emptyset$ for any $i \neq j$, so $\psi(y) = \|y - a_i\|^2 - \phi_i$ for $y \in L_i(\phi)$.

Corollary 9.2. The dual problem is equivalent to finding $\phi \in \mathbb{R}^n$ that minimizes

$$f(\phi) := \mu^T \phi + \sum_{i=1}^n \int_{L_i(\phi)} \|y - a_i\|^2 - \phi_i d\nu.$$

.

Proposition 9.3. Each Laguerre cell is a polyhedron.

Proof. We have that

$$\begin{aligned} y \in L_i(\phi) &\iff \forall j \in [n], \|y - a_i\|^2 - \phi_i \leq \|y - a_j\|^2 - \phi_j \\ &\iff \forall j \in [n], \|y\|^2 - 2y^T a_i + \|a_i\|^2 - \phi_i \leq \|y\|^2 - 2y^T a_j + \|a_j\|^2 - \phi_j \\ &\iff \forall j \in [n], 2y^T (a_i - a_j) \leq \|a_j\|^2 - \phi_j - \|a_i\|^2 + \phi_i. \end{aligned}$$

Thus $L_i(\phi)$ is an intersection of n half planes. □

A general result shows that marginalizing over some variables in a convex problem yields a convex function.

Theorem 9.4. Let f be defined as above. $f(\phi)$ is a concave function.

Proposition 9.5. Let f be defined as above. We have that

$$\frac{\partial f}{\partial \phi_i} = - \int_{L_i(\phi)} d\nu + \mu_i.$$

Proof. The μ_i is obvious. Differentiating the integral yields two terms. One term is $-\int_{L_i(\phi)} d\nu$ from differentiating the integral, and the other term is from moving the boundary of $L_i(\phi)$. However, the second term would cancel when taking the summation, since the integrands match on the boundary. □

Notice that if we can compute ∇f , we can use algorithms like gradient accent to find the maximizer of f . For instance, if $\tilde{\nu}$ is piece-wise linear on a polyhedron decomposition, then $\int_{L_i(\phi)} d\nu$ reduces to integrating an affine linear function on a polyhedron. Namely, $\int_P c + a^T x$, which is the volume of the polyhedron $\{(x, t) \in \mathbb{R}^{d+1} : x \in P, 0 \leq t \leq c + a^T x\}$.

10 Wasserstein Generative Adversarial Network

Notice that when $c(x, y) = \|x - y\|$, the twist condition is not satisfied, so the Brenier Theorem does not apply.

Lemma 10.1. [3, Lemma 6.7] For $c(x, y) = \|x - y\|$, given $X \subseteq \mathbb{R}^d$ and a bounded $\phi : X \rightarrow \mathbb{R}$, we have that ϕ^c is 1-Lipschitz.

Proof. Let $f_x(y) := \|x - y\| - \phi(x)$. We have that

$$|f_x(y_1) - f_x(y_2)| = |\|x - y_1\| - \phi(x) - \|x - y_2\| + \phi(x)| \leq \|y_1 - y_2\|.$$

Thus f_x is 1-Lipschitz. Since $\phi^c = \inf_{x \in X} f_x$, it is also 1-Lipschitz. $\square$

Lemma 10.2. [3, Lemma 6.7] For $c(x, y) = \|x - y\|$, given $X \subseteq \mathbb{R}^d$ and a 1-Lipschitz $\phi : X \rightarrow \mathbb{R}$, we have that $\phi^c = -\phi$.

Proof. We have that

$$\begin{aligned} \phi^c(y) &= \inf_{x \in X} \|x - y\| - \phi(x) \\ &\leq \|x - x\| - \phi(y) \\ &= -\phi(y). \end{aligned}$$

On the other hand, for any $x \in X$,

$$\begin{aligned} f_x(y) &= \|x - y\| - \phi(x) \\ &= \|x - y\| + (\phi(y) - \phi(x)) - \phi(y) \\ &\geq \|x - y\| - \|x - y\| - \phi(y) \\ &= -\phi(y). \end{aligned}$$

Thus, $\phi^c(y) = \inf_{x \in X} f_x(y) \geq -\phi(y)$. $\square$

Theorem 10.3 (Kantorovich-Rubenstein). For a compact $X \subseteq \mathbb{R}^d$, and $\mu, \nu \in \mathcal{P}(X)$, we have that

$$W_1(\mu, \nu) = \sup_{\eta \in Lip_1(X)} \int_X \eta d(\nu - \mu).$$

Proof. By the first Kantorovich duality theorem, we have that

$$W_1(\mu, \nu) = \sup \left\{ \int_X \phi d\mu + \int_X \psi d\nu : \phi, \psi \in C(X), \text{ such that } \phi(x) + \psi(y) \leq \|x - y\| \right\}.$$

Consider any $\phi \in C(X) = C_b(X)$, we have that $\phi^c \in Lip_1(X)$, and $\phi^{c\bar{c}} = -\phi^c$. Thus for any feasible (ϕ, ψ) , we have

$$\begin{aligned} \int_X \phi d\mu + \int_X \psi d\nu &\leq \int_X \phi d\mu + \int_X \phi^c d\nu \\ &\leq \int_X \phi^{c\bar{c}} d\mu + \int_X \phi^c d\nu \\ &= \int_X \phi^c d(\nu - \mu), \end{aligned}$$

where $\phi^c \in Lip_1(X)$. Thus,

$$\sup \left\{ \int_X \phi d\mu + \int_X \psi d\nu : \phi, \psi \in C(X), \text{ such that } \phi(x) + \psi(y) \leq \|x - y\| \right\} \leq \sup_{\eta \in Lip_1(X)} \int_X \eta d(\nu - \mu).$$

On the other hand, for any $\eta \in Lip_1(X) \subseteq C(X)$, we have that $-\eta = \eta^c \in C(X)$, and $\eta(y) + \eta^c(x) \leq \|x - y\|$, so $(-\eta, \eta)$ is feasible. Thus,

$$\sup \left\{ \int_X \phi d\mu + \int_X \psi d\nu : \phi, \psi \in C(X), \text{ such that } \phi(x) + \psi(y) \leq \|x - y\| \right\} \geq \sup_{\eta \in Lip_1(X)} \int_X \eta d(\nu - \mu).$$

□

Remark. The result is still true when X is unbounded, but μ, ν have finite first moment.

Consider the following problem: Given a probability distribution $P_r \in \mathcal{P}(X)$, where $X \subseteq \mathbb{R}^N$, only accessible via samples, and a family of probability distribution $P_\theta \in \mathcal{P}(X)$ for $\theta \in \Omega \subseteq \mathbb{R}^d$. We want to find θ such that P_θ is close to P_r . One traditional approach was to use the KL divergence $K(P_r || P_\theta)$, but it blows up to infinity if $\exists B \subseteq X$ with $P_r(B) > 0, P_\theta(B) = 0$. Another way is to use JS divergence

$$JS(\mu || \nu) := \frac{K(\mu || \chi) + K(\nu || \chi)}{2},$$

where $\chi = \frac{\mu + \nu}{2}$. However if the support of the “closest” P_θ and P_r only intersects on a measure zero set, the JS divergence would be constant.

Instead, we use the Wasserstein distance, and assume $P_\theta := (g_\theta)_\# p$ where p is a fixed probability measure on a low dimensional set Z , say the normal distribution $N(0, 1)$, and $g_\theta : Z \rightarrow X$ is a neural network in which z is the randomized input to the network, and θ is the vector of weights.

Theorem 10.4 (6.7.1 Arjovsky, Chintala, Bottou 2017). *Assuming various smoothness conditions, we have*

$$\nabla_\theta W_1(P_r, P_\theta)|_{\theta=\theta_0} = - \int_Z \nabla_\theta f_\theta(g_\theta(z))|_{\theta=\theta_0} dp,$$

where f_θ is the Kantorovich dual optimal solution for $W_1(p_r, P_\theta)$.

Proof. By [3, 6.5.3], $W_1(\mu, \nu + \epsilon\chi) - W_1(\mu, \nu) = -\epsilon \int_\phi d\chi + o(\epsilon)$ for $\chi(X) = 0$, where $(\phi, -\phi)$ is the optimal dual solution for $W_1(\mu, \nu)$. □

A generative adversarial network consists of two neural networks, the generator and the critic. The generator tries to find optimal θ that minimizes the $\int_X \eta d(P_r - P_\theta)$, and the critic tries to find $\eta \in Lip_1(X)$ that maximizes $\int_X \eta d(P_r - P_\theta)$.

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